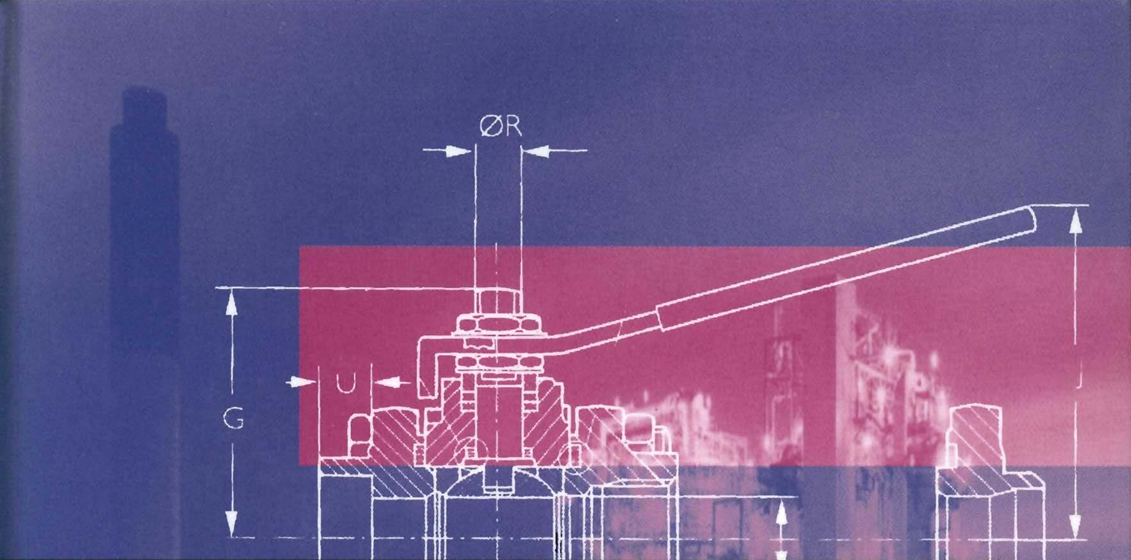

Valves, Piping & Pipelines Handbook

Third Edition

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Valves, Piping and Pipelines Handbook

3rd Edition

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T. Christopher Dickenson F.I.Mgt.

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TECHNOLOGY

UK	Elsevier Science Ltd, The Boulevard, Langford Lane, Kidlington, Oxford OX5 1GB, UK
USA	Elsevier Science Inc., 665 Avenue of the Americas, New York, NY 10010, USA
JAPAN	Elsevier Science Japan, Tsunashima Building Annex, 3-20-12 Yushima, Bunkyo-ku, Tokyo 113, Japan

Third edition 1999

Library of Congress Cataloging-in-Publication Data
Dickenson, T. Christopher.
Valves, piping, and pipelines handbook / T. Christopher Dickenson. – 3rd ed.
p. cm
ISBN 1-85617-252-X (hc)
1. Valves Handbooks, manuals, etc. 2. Piping Handbooks, manuals, etc. 3. Pipelines Handbooks, manuals, etc. I. Title.
TS277.D53 1999 99-26575
621.8′4–dc21 CIP

British Library Cataloguing in Publication Data
A catalogue record for this title is available from the British Library.

ISBN 1 85617 252X

Published by
Elsevier Advanced Technology,
The Boulevard, Langford Lane, Kidlington, Oxford OX5 1GB, UK
Tel.: +44(0) 1865 843000
Fax: +44(0) 1865 843971

Printed and bound in Great Britain by
Cambridge University Press

Preface

"A Vital Contribution to Modern Industry"

Over recent years, a number of significant developments in the application of valves have taken place: the increasing use of actuator devices, the introduction of more valve designs capable of reliable operation in difficult fluid handling situations; low noise technology and most importantly, the increasing attention being paid to product safety and reliability. Digital technology is making an impact on this market with manufacturers developing intelligent (smart) control valves incorporating control functions and interfaces.

Computer Integrated Processing is now a fact of life.

New metallic materials and coatings available make it possible to improve application ranges and reliability. New and improved polymers, plastic composite materials and ceramics are all playing their part.

Fibre-reinforced plastic pipe systems, glass-reinforced epoxy pipe systems and the traditional low-cost polyester pipe systems have all undergone sophisticated design and manufacturing technology changes. The potential for growth and expansion of the industry is huge.

The 3rd Edition of the *Valves, Piping and Pipelines Handbook* salutes these developments and provides the engineer with a timely first source of reference for the selection and application of valves and pipes.

It would not have been possible to provide so much information and data in this Handbook without the co-operation given by the individuals and companies listed overleaf, as well as the manufacturers who supplied information and data. Their contribution is gratefully acknowledged.

This is the decade of the customer, and the *Valves, Piping and Pipelines Handbook 3rd Edition* is intended to provide essential, practical product information and reference data where and when they are needed most.

T. Christopher Dickenson F.I.Mgt.
September 1999

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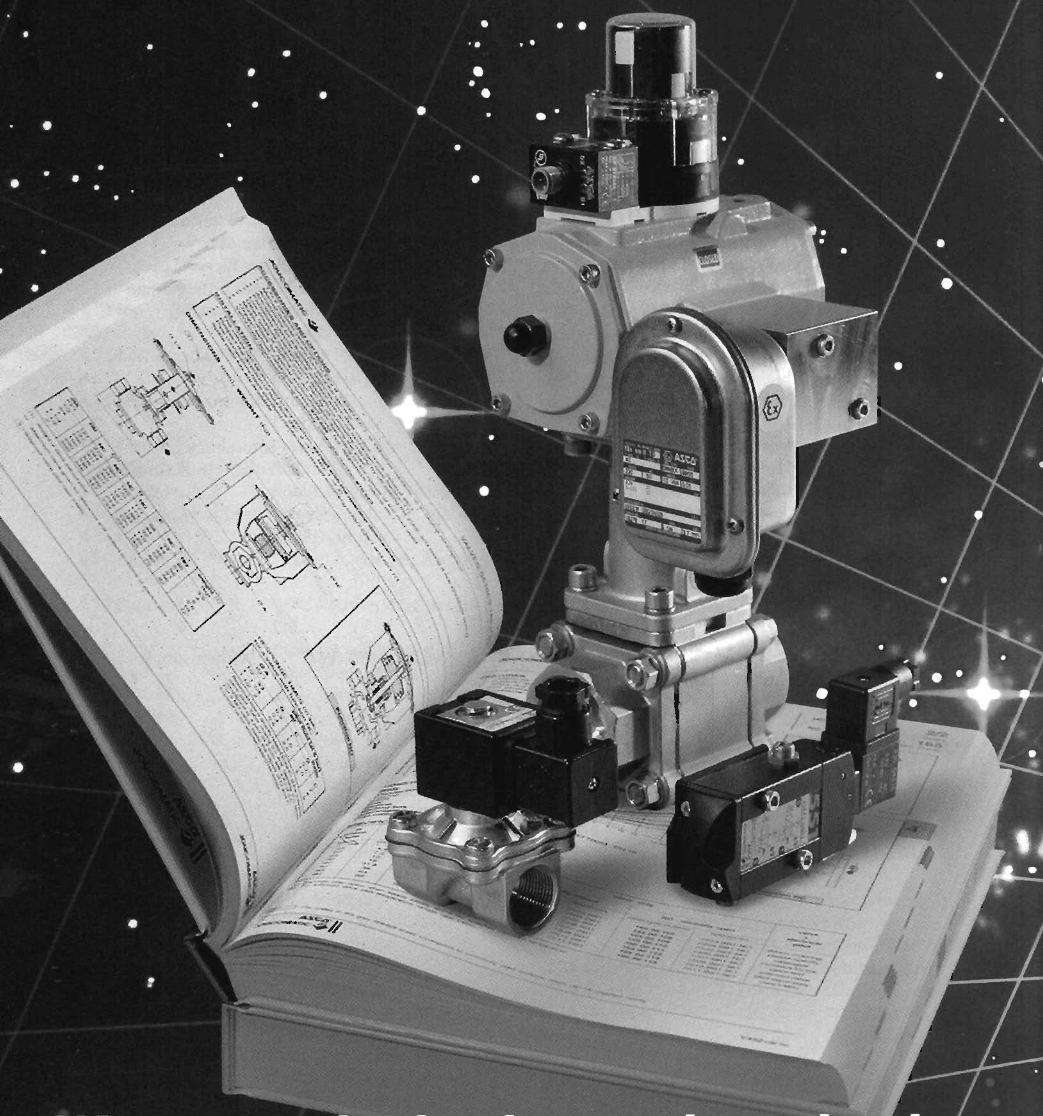


SECTION 1

Fundamentals

Classification of Valves

Basic Valve Nomenclature

Valve Selection Guides

Pipes and Pipelines—Definitions and Explanations

Classification of Valves

Valves may be classified in a number of ways, e.g. by category (general type), specific type, purpose or name; or by flow characteristics (e.g. straight-through, full-flow or throttled-flow). Descriptions can also differ slightly in different countries although the main type names are established internationally (with some exceptions).

Classification of valves by category is given in Table 1. This follows British Standards and general practice adopted by British manufacturers, but is also generally applicable to American practice. One major difference in this respect is that the important class of ball valves is considered as a type of plug valve in the tabular summary, whereas American practice would favour regarding it as a separate category. The ball valve is, in fact, a major type in its own right.

Modular ball valve based on standardised components.

Table 1. Classification of valves

Category	Patterns	Types of construction	Remarks
Cock	1. Plug 2. Gland 3. Packed cock 4. Compound gland	Tapered plug Plug retained by gland or packing Packing between plug face and body seat Stuffing box in cover	Also parallel plug.
Plug valve	1. Plain 2. Lubricated	1. Taper plug 2. Parallel plug 3. Ball plug	Passage through port in rotatable. plug supported or mounted to reduce friction.
Screw-down stop valve	1. Inside screw 2. Outside screw	1. Globe valve 2. Angle 3. Oblique 4. Others	Spherical body. Spherical body with ends at right angles. Spherical body, stem axis oblique. Usually described by type (e.g. needle valve) or geometry of body (e.g. tee valve).
Gate valve (wedge gate valve) Gate valve (slide valve)	1. Inside screw 2. Outside screw 3. Lever (a) Sliding stem (b) Rotary stem	1. Wedge (gate) 2. Sluice (valve) 3. Double disc	Closure effected by wedge action. (a) Solid wedge or (b) Split wedge. Solid wedge gate valve. Gate composed of parallel sliding discs or slides.
Check valve	1. Horizontal 2. Vertical 3. Angle	1. Swing (check) 2. Lift (a) Disc (b) Piston (c) Ball 3. Foot (valve)	Hinged flap check mechanism. Disc check mechanism. Disc plus piston check mechanism. Ball check. Check valve fitted to bottom of a suction pipe.
Butterfly valve	1. Double flanged 2. Water (a) Single flange (b) Flangeless	Each flange is individually bolted. Primarily designed for insertion between pipe flanges	Based on rotatable disc valve.
Diaphragm valve		Flexible diaphragm mounted over a weir.	
Ball (float valve)	1. Single beat 2. Double beat	1. Direct (lever)-operated 2. Pressure-operated 3. Droptight 4. Non-droptight	Single beat—flow through single seating ring. Double beat—flow through two seating rings.

Table 1 (*continued*)

Category	Patterns	Types of construction	Remarks
Safety valve	1. Direct spring-loaded 2. Direct weight-loaded 3. Lever and spring-loaded 4. Lever and weight-loaded 5. Tension spring-loaded 6. Torsion bar		Also designated by: 1. High-lift valve. 2. Full-lift valve. 3. Pilot-operated valve. 4. Electrically-assisted valve.
Relief valve	1. Direct spring-loaded 2. Direct weight-loaded		Also designated by: 1. High-lift (relief) valve. 2. Pilot-operated (relief) valve.
Pressure control valve	1. Self-contained 2. Spring-loaded 3. Weight-loaded 4. Pressure-loaded 5. Externally-piped 6. Tight-closing 7. Non-tight-closing 8. Relay-operated	1. Pressure-reducing 2. Pressure-retaining 3. Indirect	
Air relief valve	1. Single-orifice LP 2. Single-orifice HP 3. Single-orifice with integral isolating valve 4. Double-orifice with integral isolating valve		
Turbine valve	1. Regulating 2. Quick-closing 3. Starting 4. Exhaust 5. Guarding		
Free discharge valve	1. Needle-type 2. Hollow jet-type 3. Sleeve-type		

Valves are classified and described by specific type in Section 2, which also include a number of individual designs best categorised as miscellaneous. Some other valve types are given in Table 2.

Descriptions of various valve types may also differ and Table 3 lists some other descriptions, standard terminology in this case being based on that adopted for Table 1. This is by no means complete, but is offered as a general guide.

Table 2. Some other valve types

Category	Description
Flow-regulating valve	For controlling rate of flow in a system.
Temperature-regulating valve	For controlling fluid temperature level in a system.
Automatic process-control valve	For controlling rate of flow relative to value of a command system.
Anti-vacuum valve	An automatic type of air valve for the prevention of the formation of vacuum or the release of vacuum in large bore pipelines.
Blow-down valve	A valve which is used for cleaning sludge and other foreign matter from a boiler.
Bulkhead valve	A gate valve.
Free-ball valve	A valve in which a ball, free to rotate in any direction, is moved at 90° to the flow stream from a position removed from the flow stream until finally rolling into a circular orifice for shut off.
Fusible-link or fire valve	A fire prevention valve which has a weighted lever held open by a wire and fusible link which melts at an increase in room temperature.
Hydraulic valve	A control valve for either water, oil, or hydraulic systems.
Jet-dispersal valve	A valve incorporating an element by virtue of which the energy within the emitting jet is dissipated.
Penstock	A single-faced type of valve consisting of an open frame and door, and used in terminal positions only: usually located in tanks or channels as a means of controlling flow into a pipe.
Plate valve	A gate valve incorporating a sluicing effect.
Radiator valve	A valve for controlling the flow of water through a radiator.
Rotary-slide valve	A valve in which rotation of internal parts regulates flow by opening or closing a series of segmental ports.
Rotary valve	A spherical-plug valve in which the plug, which rotates through 90°, is provided with a circular waterway to match the body and ports.
Solenoid valve	A valve operated by an electrical solenoid.
Spectacle-eye valve	A type of parallel slide valve in which the 'spectacle gate' has one 'lens' of circular waterway and the other of solid section.
Thermostatic mixing valve	A valve which combines temperature selection and flow control in the same body.
Throttle valve	A non-tight-closing butterfly valve with a centrally-hinged flap which can be locked in any desired position.

Classification of valves by function yields the following general list where any individual type of valve may be capable of performing one or more of these functions. Excluded from this list are specific functions or specialised services for which special designs of valves are normally employed.

(i) On-off service.
(ii) Throttling or flow control.
(iii) Preventing of reverse flow.

Table 3. Terminology

General or 'popular' description	Standard terminology
Back-pressure valve	Check valve
Block and bleed valve	Gate valve
Clack valve	Check valve
Conduit valve	Gate valve with full-bore aperture
Controllable check valve	Screw-down stop and check valve
Controllable non-return valve	Screw-down stop and check valve
Dashpot valve	Check valve (piston-check disc-type)
Excess-flow valve	Flow-regulating valve
Excess- or minimum-pressure valve	Flow-regulating valve
Flap valve	Check valve (swing-type)
Follower-ring valve	Gate valve
Fullway valve	Gate valve
Governor valve	Pressure-control valve
Non-return self-closing valve	Check valve
Parallel-gate valve	Safety valve (direct spring-loaded)
Proportional-flow valve	Flow-regulating valve
Reflux valve	Check valve
Retention valve	Check valve
Screw-down non-return and flood valve	Screw-down stop and check valve
Wheel valve	Screw-down stop valve
Y-type valve	Oblique valve

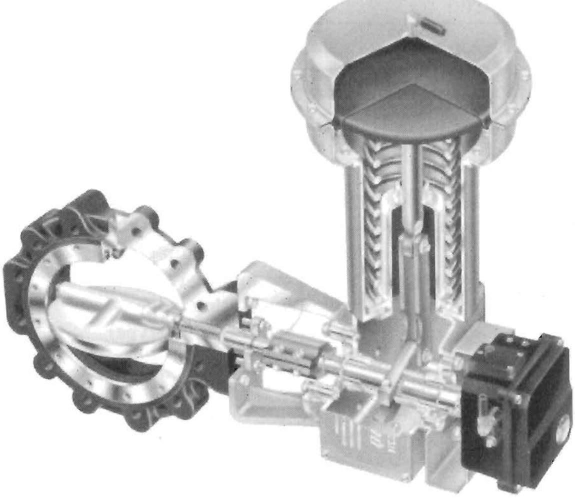
(iv) Pressure control.
(v) Directional flow control.
(vi) Sampling.
(vii) Flow limiting.

Valves classified by duty or the service they are intended to perform are described in Section 3. Necessarily these embrace types already described under specific types and the relevant chapters can be studied together where appropriate. A further source of reference and information in this respect is the chapter on Valve Selection Guides.

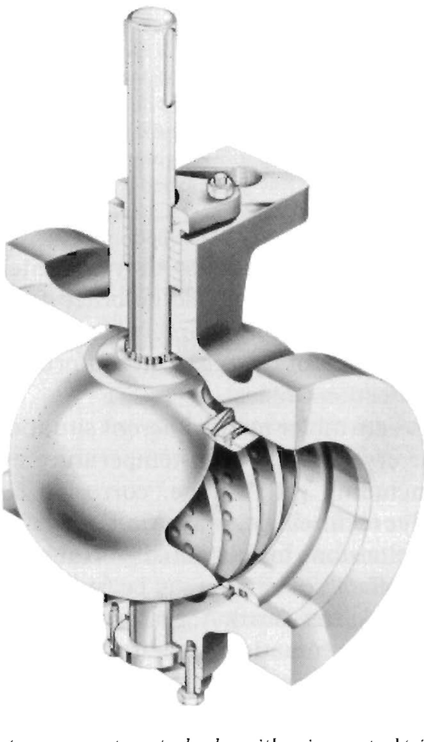
Industrial valves operate under many different situations and temperatures which range from the cryogenic to high-temperature applications and with different materials including grit, sludge, corrosive chemicals, gases and liquids. In general, valve technology is mature.

There are two main divisions in the industry: control valves giving precise control of flow and on/off valves which may be further subdivided into linear (multi-turn) and rotary (quarter-turn). Actuators which control the movement of a valve can be manual or automatic and are a major ancillary item for valves.

Valves can be purchased as standard products (commodity valves) or as engineered units, purpose-built for a specific application. The emphasis today is on providing solutions to problems and automation wherever possible.



High-performance butterfly valve with sectioned spring-diaphragm actuator for modulating control.



Rotary segment-control valve with noise-control trim.

Processes are required to be more economical and run uninterrupted for longer periods.

The intervals between production shut-downs for plant maintenance are growing longer and environmental protection legislation is becoming more stringent.

Intelligent valves, based on digital control technology and incorporating control functions and communication interfaces are already making an impact and computer integrated processing (CIP) is a reality. The most striking changes in valve technology appear to be in the field of materials of constructions with new metals, ceramics and composites being explored.

Valve connections

Valves are normally designed to take either threaded pipe ends, or with flanges for flanged connection. Threaded connections are simpler and cheaper to produce and more easily installed. However, it can prove difficult to remove valves so mounted without dismantling a considerable portion of the piping unless a number of extra fittings, such as unions, are incorporated.

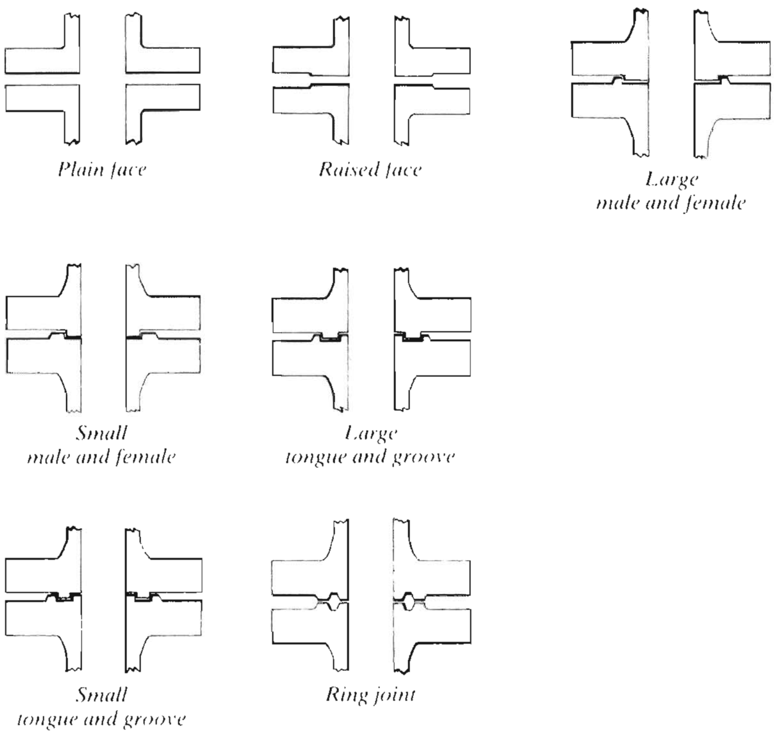


Figure 1. Flanged ends.

Flanged ends make a stronger, tighter and more leak-proof connection. Where heavy viscous media are to be controlled, as in refineries, process and chemical plants, etc., flanged-end valves are normally used. The initial cost is higher, not only because of the extra metal but because the flanges must be carefully and accurately machined. Also the installation cost is greater because companion flanges, to which the valve-end flanges are bolted, as well as gaskets, bolts and nuts must be provided.

All flat faces are commonly termed plain faces. Bronze and iron flat faces can have a machined finish. Cast iron raised faces may be smooth finished or have a serrated finish (preferably with no fewer than 16 serrations per inch) which may be spiral or concentric. Steel flat faces and raised faces should have a serrated finish of approximately 32 serrations per inch. The serrations may be either spiral or concentric.

Steel male and female and tongue-and-grooved faces should have a smooth finish. Steel ring-joint faces should have smooth finished grooves. If spiral-wound gaskets are used on flange faces, the flanges should have a smooth finish. Examples of flanged ends are shown in Figure 1.

Socket or butt-welded ends are used on all-welded pipeline systems. For specific services valves are also to permit connection to pipes by soldering or brazing. In the latter case the valve may be supplied with integral preformed brazing-material inserts.

Basic Valve Nomenclature

Most valves consist of a body containing a flow control element (discs, plug, gate, etc.) attached to and operated by rotation of a stem. (There are exceptions: e.g. swing check and pinch valves have no stem.) The stem, together with any stem seals, is enclosed within a bonnet. The top of the stem is fitted with a handwheel (or lever) for rotation of the stem (although some stems may have a sliding operation for quick action).

With threaded stems (giving a screw-down, screw-up motion) the threaded portion may be fully enclosed by the bonnet, known as inside screw; or exposed beyond the bonnet, known as outside screw. The former obviously provides maximum protection for the screw thread. Outside screws have the advantage of being easier to lubricate.

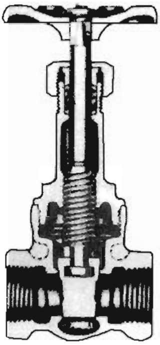
With rising-stem valves the handwheel and stem move together, giving a visual indication of the degree of valve opening. With a non-rising stem the handwheel does not rise (or fall) with the turning movement. The advantage of this type is that it can be installed in situations providing only minimum headroom above the handwheel.

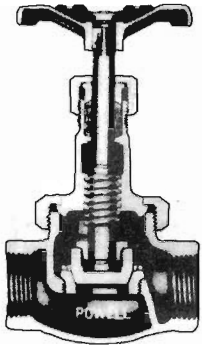
Various types of bonnet may be used, e.g. screw-in, screw-on, union-style and bolted or flanged bonnet. Screw-in or screw-on bonnets are the simplest and cheapest, but largely limited to smaller valves used on low-pressure services. Union bonnets generally provide tighter sealing and are particularly suitable for valves which are dismantled frequently for servicing. Plain (flat) flange and male- and female-flanged bonnets are generally preferred for high-temperature or high-pressure valves, and also larger sizes of valves. An alternative type for high-pressure and/or high-temperature services is the breech-lock bonnet.

Valve trim

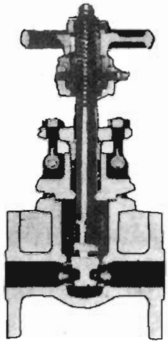
Trim is the term used to describe the parts of a valve which are replaceable, i.e. normally those parts likely to be subject to wear or degradation. The following parts are considered as trim:

Gate valves—stem, seat ring, wedge, back-seat bushing.
Globe and angle valves—stem, seat ring, disc, disc nut, back-seat bushing.


Screwed bonnet.



Union bonnet.


Bolted flanged bonnet.

Disc valve—disc, disc nut, back-seat bushing.
Swing-check valves—disc, disc holder, disc nut, side plug, carrier pin, disc-holder pin, disc-nut pin, seat rings.
Lift-check valves—disc, disc guide, seat rings.

Stem seals and other internal seals (where fitted) are arguably included under the definition of trim, but are not normally used in describing trim materials.

Standard abbreviations

The following abbreviations are used to describe or designate valve parts, features, etc.:

All iron	all parts of iron construction.
All bronze	all parts of bronze construction.
BB	bolted (flanged) bonnet.
CWp	cold working pressure.
DD	double disc.
DW	double wedge.
FE	flanged end (connection).
FF	flat flange.
IBBM	iron body bronze-mounted.
IPS	iron pipe size.
ISNRS	inside screw non-rising stem.
ISRS	inside screw rising-stem.
NRS	non-rising stem.
RF	raised flange.
RS	rising stem.
SIB	screwed bonnet.

SW	solid wedge.
S int or int S	internal seat.
S ren or ren S	renewable seat.
OS & Y	outside screw and yoke.
WOG	water, oil, gas pressure rating.

Nomenclature covering the individual parts of various different types of valves is included in Section 2.

Valve Selection Guides

The main parameters concerned in selecting a valve or valves for a typical general service are:

(i) Fluid to be handled—this will affect both type of valve and material choice for valve construction.
(ii) Functional requirements—mainly affecting choice of type of valve.
(iii) Operating conditions—affecting both choice of valve type and constructional materials.
(iv) Flow characteristics and frictional loss—where not already covered by (ii), or setting additional specific or desirable requirements.
(v) Size of valve—this again can affect choice of type of valve (e.g. very large sizes are only available in a limited range of types); and availability (matching sizes may not be available as standard production in a particular type).
(vi) Any special requirements—e.g. quick-opening, free-draining, etc.

In the case of specific services, choice of valve type may be somewhat simplified, e.g. by following established practice or selecting from valves specifically produced for that particular service.

On a broad basis, Table 1 summarises the applications of the main types of general purpose valves. It has only limited use as a selection guide, i.e. can be regarded as a starting point. Table 2 carries general selection a stage further in listing valve types normally used for specific services. Table 3 is a particularly useful expansion of the same theme relating the suitability of different valve types to specific functional requirements.

Normally, for general services (and for many specific services), several valve types may appear as possible choices. These may then need evaluating individually, and comparing on the basis of the flow characteristics they offer. Even more important, calculations may be necessary to establish a suitable size of valve to meet a specific performance requirement, e.g. a maximum acceptance pressure drop or head loss through the valve.

Table 1. Valve types—typical applications

Valve category	General application(s)	Actuation	Remarks
Screw-down stop valve	Shut-off or regulation of flow of liquids and gases (e.g. steam).	(i) Handwheel (ii) Electric motor (iii) hydraulic systems actuator (iv) Hydraulic actuator (v) Air motor	(a) Limited application for low-pressure/low-volume systems because of relatively high cost. (b) Limited suitability for handling viscous or contaminated fluids.
Cock	Low-pressure service on clean, cold fluids (e.g. water, oils, etc.).	Usually manual	Limited application for steam services.
Gate valve	Normally used either fully open or fully closed for on–off regulation on water, oil, gas, steam and other fluid services.	(i) Handwheel (ii) Electric motor (iii) hydraulic systems actuator (iv) Hydraulic actuator (v) Air motor	(a) Not recommended for use as throttling valve. (b) Solid wedge gate is free from 'chatter' and jamming.
Parallel-slide valve	Regulation of flow, particularly in main services in process industries and steam power plant.		(a) Offers unrestricted bore at full opening. (b) Can incorporate venturi bore to reduce operating torque.
Butterfly valve	Shut-off and regulation in large pipelines in waterworks, process industries, petrochemical industries, hydroelectric power stations and thermal power stations.	(i) Handwheel (ii) Electric motor (iii) hydraulic systems actuator (iv) Hydraulic actuator (v) Air motor	(a) Relatively simple construction. (b) Readily produced in very large sizes (e.g. up to 18 ft or more).
Diaphragm valve	Wide range of applications in all services for flow regulation.	(i) Handwheel (ii) Electric motor (iii) hydraulic systems actuator (iv) Hydraulic actuator (v) Air motor	(a) Can handle all types of fluids, including slurries, sludges, etc., and contaminated fluids. (b) Limited for steam services by temperature and pressure rating of diaphragm.
Ball valve	Wide range of applications for all sizes, including very large sizes in oil pipelines, etc.	(i) Handwheel (ii) Electric motor (iii) hydraulic systems actuator (iv) Hydraulic actuator	(a) Unrestricted bore at full opening. (b) Can handle all types of fluids. (c) Low operating torque. (d) Not normally used as a throttling valve.
Pinch valve	Particularly suitable for handling corrosive media, solids in suspension slurries, etc.	(i) Mechanical (ii) Electric motor (iii) hydraulic systems actuator (iv) Hydraulic actuator (v) Fluid pressure (modified design)	(a) Unrestricted bore at full opening. (b) Can handle all types of fluids. (c) Simple servicing. (d) Limited maximum pressure rating.
Automatic process-control valve	Designed to meet particular service conditions.	To meet particular service conditions	Most commonly of single or double beat-globe valve configuration.
Air-relief valve	Used in water works, etc., to release entrapped air and prevent formation of vacuum 'pockets'	Automatic—responding to changes in flow pressure	
Turbine valve	Designed to meet requirements of steam and water turbines in industrial, marine and power	To meet particular service conditions	Provides guaranteed control over maximum and minimum turbine speeds and power in association

Table 2. Valve types for specific services

Service	Main	Secondary
Gases	Butterfly valves Check valves Diaphragm valves Lubricated plug valves Screw-down stop valves	Pressure-control valves Pressure-relief valves Pressure-reducing valves Safety valves Relief valves
Liquids, clear up to sludges and sewage	Butterfly valves Screw-down stop valves Gate valves Lubricated plug valves Diaphragm valves Pinch valves	
Slurries and liquids heavily contaminated with solids	Butterfly valves Pinch valves Gate valves Screw-down stop valves Lubricated plug valves	
Steam	Butterfly valves Gate valves Screw-down stop valves Turbine valves	Check valves Pressure-control valves Pre-superheated valves Safety and relief valves

Valve coefficients and flow values

The valve coefficient is a convenient method of relating flow rates to pressure drop through valves and, in fact, is sometimes called a flow value. This coefficient can only be determined empirically for a specific type of valve as it will be influenced by detail design and construction. It will also vary with the physical size of the valve and the degree of opening in the valve. Valve coefficient values are normally quoted for 100% opening (full open), with individual valves for each size.

Some confusion can arise from the fact that the coefficient quoted for a valve can have three different values depending on the basic units on which it was computed. Normally these are apparent from the designation of the valve coefficient, viz:

C_v	in units of US gal/min, $1bf/in^2$
K_v or k_v	in units of l/min, bar
f	in Imperial units of Imp gal/min, $1bf/in^2$

The following conversions apply:

	K_v	C_v	f
K_v	—	14.28	17.09
C_v	0.07	—	1.1966
f	0.0589	0.8357	—

Table 3. Typical valve suitability chart

Valve type	Service or function										
	On–off	Throttling	Diverting	No reverse flow	Pressure control	Flow control	Pressure relief	Quick opening	Free draining	Low pressure drop	Handling solids in suspension
Ball	S	M	S	—	—	—	—	S	—	S	LS
Butterfly	S	S	—	—	—	S	—	S	S	S	S
Diaphragm	S	M	—	—	—	—	—	M	M	—	S
Gate	S	—	—	—	—	—	—	S	S	S	—
Globe	S	M	—	—	—	M	—	—	—	—	—
Plug	S	M	S	—	—	M	—	S	S	S	LS
Oblique (Y)	S	M	—	—	—	M	—	—	—	—	—
Pinch	S	S	—	—	—	S	—	—	S	S	S
Slide	—	M	—	—	—	M	—	M	S	S	S
Swing-check	—	—	—	S	—	—	—	—	—	S	—
Tilting-disc	—	—	—	S	—	—	—	—	—	S	—
Lift-check	—	—	—	S	—	—	—	—	—	—	—
Piston-check	—	—	—	S	—	—	—	—	—	—	—
Butterfly-check	—	—	—	S	—	—	—	—	—	—	—
Pressure-relief	S	—	—	—	—	—	S	—	—	—	—
Pressure-reducing	—	—	—	—	S	—	—	—	—	—	—
Sampling	S	—	—	—	—	—	—	—	—	—	—
Needle	—	S	—	—	—	—	—	—	—	—	—

Key:

S = Suitable choice

M = May be suitable in modified form

LS = Limited suitability

Typical flow coefficient equations can be shown as follows:

For liquids

$$Q = C_v \sqrt{\frac{\Delta P}{(s.g.)}}$$

where

Q = flow, gallons per minute
C_v = flow coefficient
ΔP = pressure drop, psi
s.g. = specific gravity (water=1)

For gases (non-critical flow):

$$Q = 16.07\, C_v \sqrt{\frac{Z(P_1^2 - P_2^2)}{T \times (s.g.)}}$$

where

Q = flow, SCFM
C_v = flow coefficient
P_1 = upstream pressure, psia
P_2 = downstream pressure, psia
Z = compressibility factor
T = absolute temperature (°F+460)
s.g. = specific gravity (air=1)

For gases (critical flow):

$$P_2/P_1 = R, \text{ and}$$

$$Q = 16.07\, C_v P_1 J \sqrt{\frac{Z}{T \times (s.g.)}}$$

where R and J are functions of the specific heat ratio 'r' as follows:

r	R	J	r	R	J
1.20	0.564	0.825	1.36	0.535	0.845
1.22	0.561	0.828	1.38	0.532	0.847
1.24	0.557	0.831	1.40	0.528	0.849
1.26	0.553	0.833	1.42	0.524	0.851
1.28	0.549	0.836	1.44	0.521	0.853
1.30	0.546	0.838	1.46	0.528	0.855
1.32	0.542	0.840	1.48	0.515	0.857
1.34	0.539	0.843	1.50	0.512	0.859

Combining flow coefficients

(1) Flow in parallel: $C_v = C_{v1} + C_{v2} + C_{v3} + \ldots$

(2) Flow in series: $\left(\frac{1}{C_v}\right)^2 = \left(\frac{1}{C_{v1}}\right)^3 + \left(\frac{1}{C_{v2}}\right)^2 + \left(\frac{1}{C_{v3}}\right)^2 + \ldots$

Flow characteristics of valves

Where the flow characteristics through the valve are of significance, the following notes can be useful.

Plug valves (Figure 1) offer a straightway passage through the ports with a minimum of turbulence. Flow can be in either direction and a quarter-turn will fully open or fully close the valve. Similar comment applies to ball valves.

Gate valves (Figure 2) present a substantially straightway flow through the ports in the full-open position since the wedge or 'gate' is lifted clear of the flow passage. Turbulence and pressure drop are low. Again flow can be in either direction.

Globe valves (Figure 3) are normally installed so that pressure is under the disc, assisting operation and eliminating a certain amount of erosive action. Turbulence and pressure drop are higher than with straightway valves.

Angle valves (Figure 4) have similar characteristics to globe valves, with flow directed through 90°. Again flow is normally directed under the disc. Reverse flow may be used in the case of high-temperature steam. Ball, globe and angle valves are suitable for throttling.

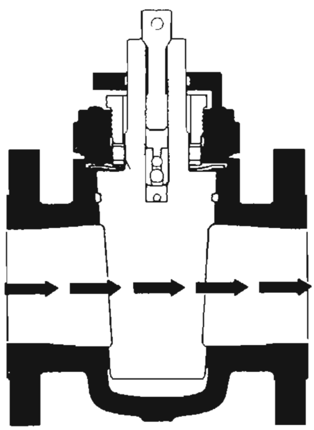


Figure 1. Plug valve.

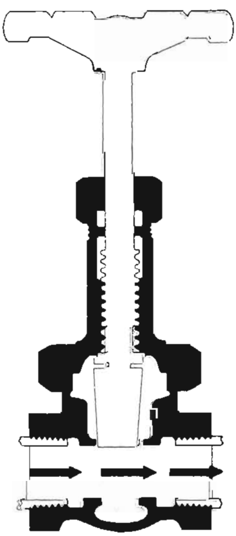
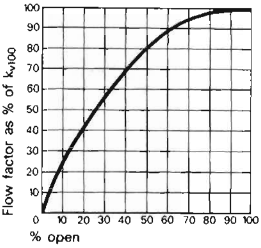


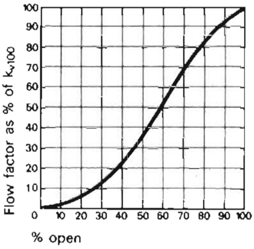
Figure 2. Gate valve.

Nominal Size	K_{v100}
3/8	41
1/2	95
3/4	180
1	327
1 1/4	484
1 1/2	725
2	1130
2 1/2	1700
3	-



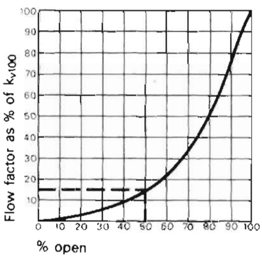
Gate

Nominal Size	K_{v100}
3/8	49
1/2	77
3/4	146
1	260
1 1/4	437



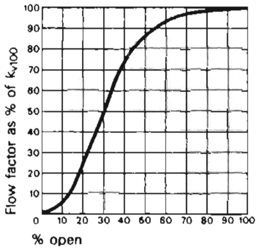
Lubricated plug

Nominal Size	K_{v100}
1/8	22
1/4	32
3/8	71
1/2	185
3/4	350
1	700
1 1/4	1000
1 1/2	1600
2	3100
3	6500
4	11000



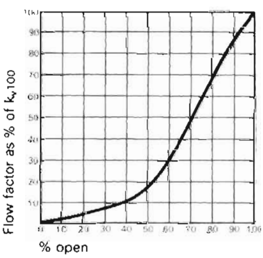
Ball

Nominal Size	K_{v100}
1/2	63
3/4	121
1	187
1 1/4	332
1 1/2	416
2	704
3	1700
4	2700



Diaphragm

Nominal Size	K_{v100}
2 1/2	1500
3	3000
4	5000
5	8000
6	12000
8	17000



Butterfly

Figure 8. Examples of flow values K_v.

The K_v table for angle-seat valves gives a K_{v100} factor of 327 for size 1 in, 484 for size $1^1/_4$ in and 725 for size $1^1/_2$ in.

In this example the correct size to use is $1^1/_4$ in (See also Table 5).

Example 2 (Figures 9 and 10)

(i) What is the K_v factor for a $1^1/_4$ in water pipeline with a flow of 300 l/min, an inlet pressure of 0.5 bar and an outlet pressure of 0 bar?

(ii) If a valve has to be fitted and the minimum acceptable flow rate in the pipeline is 250 l/min, which type of valve should be used?

Table 4. Typical 'K' values and pressure drops for various 150 mm (6 in) bore valves

Service	K value	Pressure drop*	
		bar	lbf/in^2
Globe	5.0	0.59	8.5
Swing-check	3.5	0.40	5.9
Y-pattern	2.9	0.34	4.9
Angle (globe)	2.2	0.25	3.7
Venturi parallel-slide (with eyepiece)	1.1	0.13	1.9
Butterfly	1.0	0.12	1.7
Parallel-slide without eyepiece	0.15	0.021	0.3
Parallel-slide with eyepiece	0.05	0.007	0.1
Ball (full bore)	0.05	0.007	0.1
Straight pipe (the length of an average 6 in bore valve)	0.045	0.005	0.075

*Flow 40 m/s (140 ft/s) at 24 bar (350 lbf/in^2) saturated steam.

Solution to (i) (Figure 9)

Calculate the K_v for the pipeline (K_{vp}).

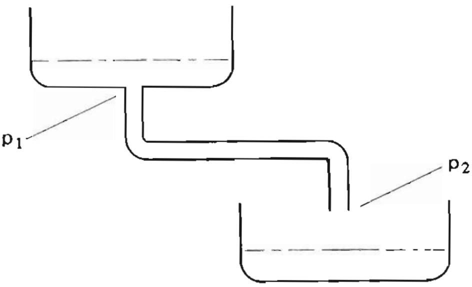


Figure 9.

Given:

$$Q = 300\ l/min$$
$$\gamma = 1\ kg/dm^3$$
$$\Delta P = P_1 - P_2 = 0.5 - 0 = 0.5\ bar$$

Then:

$$K_{vp} = Q\sqrt{\frac{\gamma}{\Delta P}}$$

$$= 300\sqrt{\frac{1}{0.5}}$$

$$K_{vp} = 424$$

Table 5. Typical sizes and operating ranges of valves

Valve	Size		Pressure range		Temperature range	
	Min.	Max.	Min.	Max.	Min.	Max.
	mm (in)		bar (lb/in^2)		°C (°F)	
Ball	6 ($^1/_4$)	1220 (48)	A	525 (7500)	−55 (−65)	300 (575)
Butterfly	50 (2)	1830 (72)	V	84 (1200)	−30 (−20)	538 (1000)
Butterfly-check	25 (1)	1830 (72)	A	84 (1200)	−18 (0)	260 (500)
Gate	3 ($^1/_8$)	1220 (48)	V	700 (10,000)	−277 (−455)	675 (1250)
Globe	3 ($^1/_8$)	760 (30)	V	700 (10,000)	−272 (−455)	540 (1000)
Plug, lubricated	6 ($^1/_4$)	760 (30)	A	350 (5000)	−40 (−40)	315 (600)
Plug, non-lubricated	6 ($^1/_4$)	406 (16)	A	210 (3000)	−75 (−100)	220 (425)
Swing-check	6 ($^1/_4$)	610 (24)	A	175 (2500)	−18 (0)	540 (1200)
Swing-check ,Y-type	6 ($^1/_4$)	150 (6)	A	175 (2500)	−18 (0)	540 (1200)
Lift-check	6 ($^1/_4$)	250 (10)	A	175 (2500)	−18 (0)	540 (1200)
Tilting-disc	50 (2)	760 (30)	A	84 (1200)	−260 (−450)	590 (1100)
Diaphragm	3 ($^1/_8$)	610 (24)	V	21 (300)	−50 (−60)	230 (450)
Y (oblique)	3 ($^1/_8$)	760 (30)	V	175 (2500)	−272 (455)	540 (1000)
Slide	50 (2)	1900 (75)	A	28 (400)	−18 (0)	650 (1200)
Pinch	25 (1)	305 (12)	V	21 (300)	−75 (−100)	260 (500)
Needle	3 ($^1/_8$)	25 (1)	V	700 (10,000)	−78 (−100)	260 (500)

Key:
A = Atmospheric
V = Vacuum.

Solution to (ii) (Figure 10)

First it is necessary to calculate the K_v factor for the total system (K_{vt}).

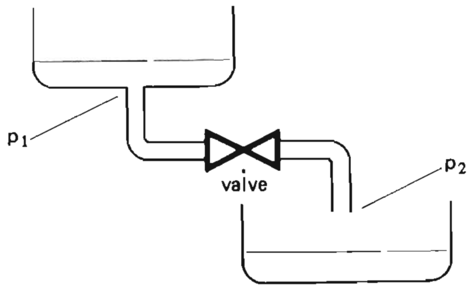


Figure 10.

Given:

$$Q = 250\ \text{l/min}$$
$$\gamma = 1\ \text{kg/dm}^3$$
$$\Delta P = P_1 - P_2 = 0.5 - 0 = 0.5\ \text{bar}$$

Then:

$$K_{vt} = Q\sqrt{\frac{1}{\Delta P}}$$

$$= 250\sqrt{\frac{1}{0.5}}$$

$$K_{vt} = 354$$

The K_v factor for the valve (K_{vv}) can now be established by subtracting the K_v factor for the pipeline (K_{vp}) from the k_v factor for the total system (K_{vt}). For this purpose, the formula for calculating the flow factors in series should be used, which is:

$$\frac{1}{K_v^2 x} = \frac{1}{K_{v1}^2} + \frac{1}{K_{v2}^2} + \ldots \frac{1}{K_{vn}^2}$$

thus:

$$\frac{1}{K_{vt}^2} = \frac{1}{K_{vv}^2} + \frac{1}{K_{vp}^2}$$

$$\frac{1}{K_{vv}^2} = \frac{1}{K_{vt}^2} - \frac{1}{K_{vp}^2}$$

$$\frac{1}{K_{vv}^2} = \frac{1}{354^2} - \frac{1}{424^2}$$

$$= 7.98 \times 10^{-6} - 5.56 \times 10^{-6}$$

$$= 2.42 \times 10^{-6}$$

$$K_{vv} = \sqrt{\frac{1}{2.42 \times 10^{-6}}}$$

$$= 643$$

The calculation shows that the valve used must be one with a minimum K_{V100} factor of 640.

From the K_V tables it can be seen that a $1^1/_4$ in ball valve has a K_{V100} factor of 1000 and a $1^1/_4$ in diaphragm valve has a K_{V100} factor of 332.

Therefore only the $1^1/_4$ in ball valve can be used.

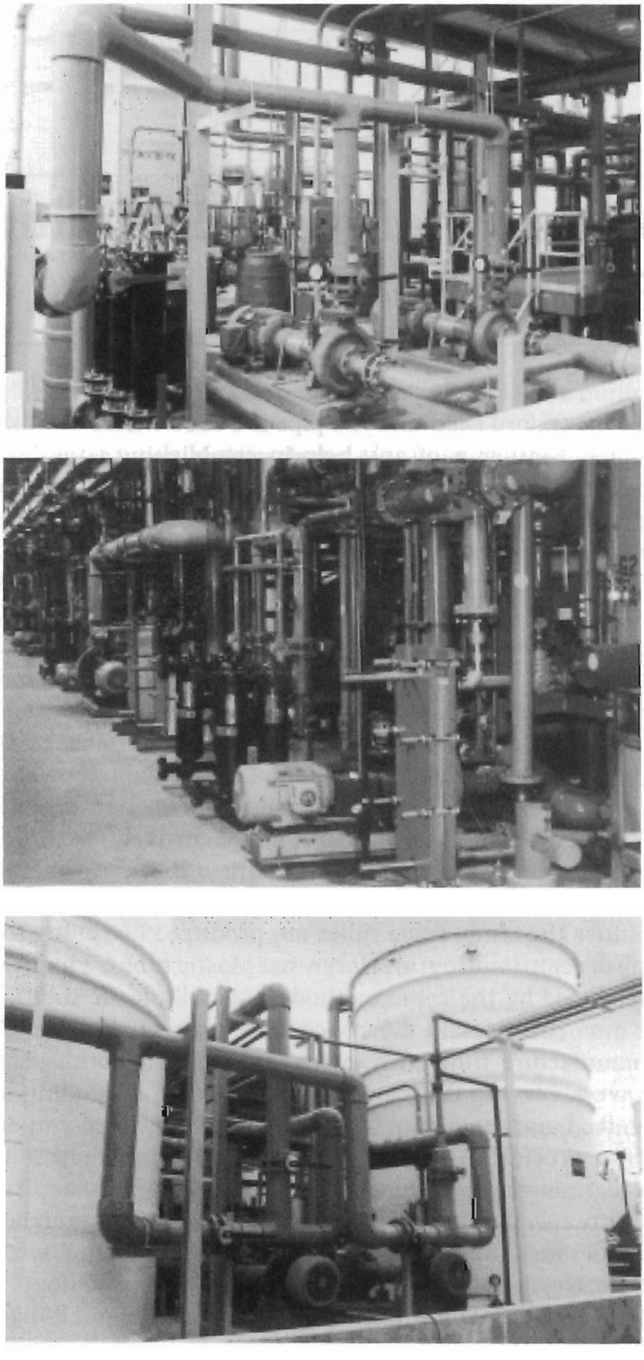
Pipes and Pipelines—Definitions and Explanations

According to the Oxford Dictionary, a pipe is a tube whereas a tube is a long, hollow cylinder. Neither is of any help in establishing true definitions, for there are recognised differences between pipes and tubes—but not those the dictionary gives. The more obvious distinction is that 'a pipe is a big tube, and a tube is a small pipe'—which is not far from the truth in application. But we are also concerned with differences in usage of terms in different industries—and in different countries.

Taking the big tube/small pipe premise as substantially correct, we can further comment that pipes which may run up to several metres or feet in diameter are cast, spun, welded up or otherwise fabricated, depending on the materials and sizes involved. Nobody could logically visualise producing very small sizes of pipes—e.g. under 25 mm (1 in) diameter—by such time consuming methods. It is much quicker and cheaper to produce them by extrusion. Hence tubes are basically (but not exclusively) extruded products, involving reduction in size during manufacture in the case of metal tubes, and a moulding process in the case of plastic tubes.

Just to confuse the issue, some tubes are produced by rolling to shape and seam welding or seam jointing; and large-size plastic tubes, which then become pipes, are produced by the same methods as small plastic tubes. But ignore that for the moment. A main difference does emerge from the two different methods of manufacture. Inherently, tubes have a smooth bore as manufactured. Pipes will have a varying degree of bore roughness, depending both on the material involved and the actual fabrication method. Once you extend tube-manufacturing process to pipe production, then these pipes also have a smooth bore (e.g. plastic pipes). Pipes produced by pipe-manufacturing methods normally require specific after-treatment to render them smooth bore.

With this difference (and there are exceptions to the rule), we can further differentiate between the two by size ranges and terminology adopted by different industries. One of the main users of smooth-bore small-diameter tubes is the hydraulic industry where line sizes may range from 3 mm (1/8 in) bore up to 32 mm (13 in) bore, or larger in low-pressure hydraulic systems—and



CPVC pipes and tubes.

we have called them lines, not pipes or tubes. The industry itself may call them hydraulic pipes, hydraulic tubes or hydraulic lines; and larger hydraulic tubes (pipe sizes!) are produced for cylinder tubes.

Industries and applications concerned with the conveyance of fluid products almost invariably refer to their tubular products as pipes or piping. Again, sizes may range down into tube sizes (and even be drawn or extruded products or true tubes)—e.g. gas pipes and small-bore water service pipes. But they are still pipes or piping. And the system they provide is a pipeline.

Hopefully this has established a satisfactory definition and explanation of why the title of this handbook is specifically concerned with pipes and pipelines, for these are the areas mainly covered. And those who work in these areas call

Ductile iron pipe for drinking water applications.

Summary of pipe materials—metallic

Material	Manufacturing process	Size range	Typical applications	Remarks
Aluminium	Drawing or rolling (seamless tube)		Cryogenic and chemical pipelines; lightweight hydraulic pipes.	Low weight and good corrosion resistance.
Copper	Drawing or rolling (seamless tubing)	Mainly small bore tubes	Marine applications. Hot water services (domestic).	Resistant to corrosion but costly.
Ductile iron	Spinning	Up to 600 mm (24 in)	Gas and water distribution systems.	Stronger than cast iron.
Grey cast iron	Casting	Up to 1200 mm (48 in)	Gas, water and drainage systems.	Brittle material.
Malleable iron	Heat-treated casting		Mainly used for small fittings.	Less brittle than cast iron.
Steel	Various	Up to 4000 mm (160 in)	Gas and oil pipelines.	Available in a wide range of tensile strengths.
Stainless steel	Various		Cryogenic and chemical pipelines. Stainless steel tubing for domestic water supplies, plumbing and heating.	Corrosion-resistant, but high cost.
Tungsten	Extrusion	Mainly small bore tubes	Marine applications. Specialised hydraulic systems.	Corrosion-resistant, non-sparking material.

their tubular products pipes, but tubes are mentioned and described where appropriate.

There remains one distinction between British and American practice to clarify. In the UK the handling and installation of pipes, performance calculations, etc., embracing the complete system are commonly referred to as pipework, e.g. pipework installations, pipework calculations, etc. In the USA the word 'pipework' does not appear to be accepted and is seldom, if ever, used. In the interest of rationalisation, this handbook uses the single description pipeline. It means the same as pipework.

It is to be regretted that similar rationalisation is not possible between British and American and metric units and standards. This leads to differences in values of 'flow loss' coefficients for pipe bends, valves, etc., the British/American coefficient being based on m^3/h at 1 bar pressure loss.

Equally, pipe sizes are standard in both millimetre and inch sizes, together with match fittings and valves. There are no exact equivalents. You work in standard manufactured sizes, either in millimetres or inches. To give equivalent sizes in tabular data for either would be meaningless. With rare exceptions, the exact equivalent size is just not obtainable.

That is a problem, too, which complicates the presentation of working formulae. We have attempted, within reason, to cover most possibilities in the case of the main formulae for flow-performance calculation in other forms embracing all the units most likely to be used, both in metric and Imperial units. Here, in fact, Imperial units are often less rational than their metric equivalents, with volumes expressed in cubic inches, US gallons, Imperial

Summary of pipe materials—non-metallic

Material	Size range	Corrosion resistance	Typical applications	Remarks
Asbestos cement	50 to 1050 mm (2 to 42 in)	Very good in most soils.	Buried water pipelines and drainage systems.	Brittle material.
Clay		Very good.	Drainage pipelines and ducts.	Brittle material. Normally salt-glazed.
Concrete	150 to 1950 mm (6 to 76 in)	Very good in most soils.	Drainage pipelines.	Produced in unreinforced and reinforced forms.
Spun concrete		Good resistance to sulphate attack and sewer gas.	Sewerage, drainage, etc.	Smooth, concentric bore. Smooth external finish. High-density, steel-reinforced.
Pre-stressed concrete	Up to 3000 mm (120 in)	Very good in most soils.	Large water and drainage pipelines.	Suitable for very large diameters.
Pitch/fibre	50 to 225 mm (2 to 9 in)	Very good in most soils.	Small drainage pipelines.	
Plastic pipes				
ABS	12 to 150 mm ($^1/_2$ to 6 in)	Corrosion-free but lower chemical resistance than PVC.	Alternative to PVC where better mechanical pipelines required.	Suitable for solvent jointing.
GRP	Up to 4800 mm (190 in)	Corrosion-free.	Large water and drainage pipelines.	Thermoset material. Also available in other reinforced-plastic matrix (RPM) constructions. Disadvantage: high cost.
Polyvinyl chloride UPVC	Up to 1050 mm (42 in)	Corrosion-free.	General purpose pipelines suitable for a wide variety of exterior and interior applications.	Unplasticised PVC. Suitable for solvent welding. Widely available.
Polyvinyl chloride CPVC	Up to 360 mm (14 in)	Corrosion-free.	Cold and hot water services, domestic plumbing, etc.	Rigid PVC.
Polypropylene (PP)	Up to 1000 mm (40 in) [UK sizes up to 16 in]	Similar to PE, but superior for resistance to detergents.	Applications requiring good combined temperature/pressure pipelines, e.g. effluent, pulp mills, etc.	Subject to embrittlement at low temperatures.
Polypropylene	Up to 300 mm (12 in)			Copolymer of PP with better resistance.
Polypropylene (PVDF)		High chemical-resistance, including: acids, alkalis and hydrocarbons.	Specialised applications: higher service temperatures than possible with other thermoplastic pipes.	Fusion-jointed.
Polybutylene (PB)	Up to 600 mm (24 in)		Hot-water applications—suitable for temperatures up to 110°C (230°F).	Cheaper than PEX, better abrasion resistance than PEH (particularly at elevated temperatures). Cannot be solvent-welded.
Polythene (PE) (PEL)	Up to 200 mm (8 in)	Corrosion-free.	Agriculture and irrigation.	Low-density polythene: pressures to 6 bar.
Polythene (PEM)	Up to 500 mm (20 in)	Corrosion-free.	Gas distribution. General purpose pipelines for exterior and interior applications.	Medium-density polythene: fusion-jointed or mechanical joints.
Polythene (PEH)	Up to 1800 mm (70 in)	Corrosion-free.	Water distribution, sewage, industrial effluent, etc. Gas distribution.	High-density polythene: fusion-jointed (socket, butt or saddle); also mechanical forms.
Polythene (HMW-PEH)	Up to 1200 mm (48 in)	Corrosion-free.		High molecular weight PEH: limited availability in pipe forms and expensive.
PEX			Hot water applications.	Cross-linked PE.
Fluorocarbon (FEP, PFA, PTFE)		Outstanding.	Used as liners bonded to GRP or metallic pipes for complete corrosion resistance. Limited availability in tube form (FEP and PFA).	

Equivalent specifications for stainless and high-resistant steels

Description	BS970 EN no.	AISI type	Avesta	Swedish: Fagersta	Swedish: Nyby	Sandvikan	Uddeholm	French: Government	German: Ugine	Krupps
12/14% chromium Low carbon	331S42 (56A)	410	393	R.R.J.10	1410	2.C.27	S/S.1	Z.12.C.13	FIA/FIB	V13F
12/14% chromium 15% carbon	420S29 (56B)	410	393H	R.R.J.11	1415	4.C.27	S/S.31		FIU12	V5M
12/14% chromium 35% carbon	(56D)	420	739H	R.R.S.72	1435	7.C.27	S/S.6	Z.36.C.13	S12	V3M
18/20% chromium 2% nickel	431S29 (57)	431	249EH			4N2C36	S/S.22	Z.15.CN	18–02	V1M
18% chromium		430	249	R.R.M.20	1710	I.C.36	S/S.2 2.C.34		F17	V17F Extra
26% chromium 5% nickel			453E	R.R.V.62	27–4	I.R.8	S/S.45			V.2.A. Supra Special
26% chromium 5% nickel 1.5% molybdenum			453S	R.R.V.64	27–5MO	I.R.10	S/S.44			V.8.A. Supra Special
12% chromium 12% nickel	(58D)		832P	R.R.N.J.39	14–12	2.R.I.	S/S.33		Inoxargent	V.12.A. Supra
18% chromium 8% nickel 0.08% carbon	301S15 (58$^3/_8$)	304	832M	R.I.M.291	18–8EL	O.R.2	S/S.3.M.M.	Z.5.CN. 18–08	N.S.22.S.	V.2.A. Supra

18% chromium 8% nickel	320S25 (58A)	302	832	R.R.N.J.32	18–08	2.R.2.	S/S.3	Z.10.CN. 18–08		V.2.A Normal
18% chromium 8% nickel Free machining	303S21 (58M)	303	832C			2.R.2.A.	S/S.43			
18% chromium 10% nickel 1.5% molybdenum	315S16 (58H)		832SV	R.R.N.J.41	18–8EMO	O.R.3	S/S.4.M.M.			
18% chromium 8% nickel 1.5% molybdenum	315S16 (58H)		832S	R.R.N.J.40	18–81MO	2.R.3	S/S.4			V.8.A. Normal
18% chromium 8/10% nickel 1.5% molybdenum	321S12 (58B)	321	832T	R.R.N.J.51	18–8T	I.R.4	S/S.53	Z.10.CNT. 18–08	N.S.20C	V.2.A. Extra
18% chromium 10/12% nickel 1% niobium	347S17 (58G)	347	832T			I.R.41				
17% chromium 20% nickel			254	R.R.T.80	20–20	2.R.6.	S/S.15			
25% chromium 20% nickel		310	254E	R.R.T.83	25–20	3.R.9.	S/S.25	Z.20.CNS 25/20	N.S.30	N.C.T.3
18% chromium 10% nickel 2.5% molybdenum	304S15 (58E)	316	832SK	R.R.N.J.44	18–20MO	O.R.11	S/S.24	Z.8.CND. 18–08	N.S.M.C.	V.4.A. Supra

gallons or barrels, for example, depending on the industry or application involved.

In other more specialised cases, solutions and formulae are presented in one set of units only, being those most generally used, or in which the original solutions were derived. In that case conversion tables will be necessary if you want to use these with different units entered. As a final comment here, do remember that g or gravitational acceleration is the same in Imperial or metric units—$32.2 \text{ft/s}^2 = 9.81 \text{ m/s}^2 = g$.

The following table lists ASTM (American) pipe specifications and grades with British Standard equivalents and basic material descriptions.

Pipe specifications: American and British Standards

ASTM	Material	BS equivalent
A120	Carbon steel	1387
A53 Gr.A	Carbon steel	3601/23
A53 Gr.B	Carbon steel	3601/27
A106 Gr.A	Carbon steel	3602/23
API 5L Gr.A	Carbon steel	3602/27
A106 Gr.B	Carbon steel	3602/27
API 5L Gr.B	Carbon steel	3602/27
A333 Gr.1	Killed carbon steel	3603/LT50
A333 Gr.3	3.5% nickel	3603/503LT100
A335 Gr.P1	$^1/_2$% molybdenum	3604/240
A335 Gr.P12	1%Cr $^1/_2$% Mo	3604/620
A335 Gr.P11	$1^1/_4$%Cr $^1/_2$% Mo	3604/621
A335 Gr.P22	$2^1/_4$%Cr 1% Mo	3604/622
A335 Gr.P5	5%Cr $^1/_2$% Mo	3604/625
A335 Gr.P7	7%Cr $^1/_2$% Mo	3604/627
A335 Gr.P9	9%Cr 1% Mo	3604/629
A312 Gr.Tp304	Austenitic chromium nickel	3605/304 S18 (EN58E)
A312 Gr.Tp304L	Austenitic chromium nickel (extra low carbon)	3605/304 S14
A312 Gr.Tp316	Austenitic chromium nickel molybdenum bearing	3605/316 S18 (EN58J)
A312 Gr.Tp316L	Austenitic chromium nickel molybdenum bearing (extra low carbon)	3605/316 S14
A312 Gr.Tp321	Austenitic chromium nickel titanium stabilised	3605/321 S18 (EN58B)
A312 Gr.Tp347	Austenitic chromium nickel niobium stabilised	3605/347 S18 (EN58G)

British, American and German equivalent steel specifications

BS 970	Type of steel	SAE	AISI	Werkstoff	DIN
070 M 20	'20' C steel (hot rolled or normalised)	1020	C1020	0402	C22
080 M 30	'30' C steel	1030	C1030	0501	C35
	Bright C steel	1035	C1035	0503	C45
080 A 40	'40' C steel	1040	C1040	0503	C45
070 M 55	'55' C steel	1055	C1055	0601	C60
526 M 60	'60' C–Cr steel	5160	5160	8161	58 Cr-V4
150 M 28	C–Mn steel	1027 1330	C1027 1330	5066	30 Mn 4
530 A 40	1% Cr steel	5140	5140	7035	41 Cr 4
530 A 32	1% Cr steel	5132	5132	7035	34 Cr 4
530 A 36	1% Cr steel	5135	5135	7034	37 Cr 4
530 A 40	1% Cr steel	5140	5140	7035	41 Cr 4
709 M 40	1% Cr–Mo steel	4140	4140	7220	34 Cr–Mo 4
708 M 40	Cr–Mo steel	4140	4140	7225	42 Cr–Mo 4
708 A 42	1% Cr–Mo steel	4140	4140	7225	42 Cr–Mo 4
653 M 31	3% Ni–Cr steel			5755	22 (31) Ni–Cr 41
817 M 40	1.5% Ni–Cr–Mo steel	4340	4340	6582	34 Cr–Ni–Mo 6
410 S 21	Cr-rust-resisting steel	51410	410	4006	×10 Cr 13
420 S 29	Cr-rust-resisting steel	51410	410	4021	×20 Cr 13
420 S 37	Cr-rust-resisting steel	51420	420	4021	C 20 Cr 13
420 S 45	Cr-rust-resisting steel	51420	420	4034	×40 Cr 13
420 S 21	Cr-rust-resisting steel	51416	416	4024	×20 Cr 13
		51416 Se (S1416 Se)	416 Se		
816 M 40	Low Ni–Cr–Mo steel			6582	34 Cr–Ni–Mo 6

Colour codes for pipeline identification

Originally pipes or sections of pipes were painted in colours for identification. Identification colours are now more commonly applied with bands of self adhesive tapes, with colour-fast resistance to washing down, heat, etc.

Colour coding employed in UK practice is based on BS 1710: 1960, BS 1710: 1971 and BS 1710: 1975. British Standard colours are shown with colour specifications in accordance with BS 4800.

BS 1710: 1984 Optional colour code indications for general building services

Pipe contents	Basic colour	Colour code indication
Water		
Drinking	Green	Auxiliary blue
Cooling (primary)	Green	White
Boiler feed	Green	Crimson/white/crimson
Condensate	Green	Crimson/emerald green/crimson
Chilled	Green	White/emerald green/white
Central heating ‹100°C	Green	Blue/crimson/blue
Central heating ‹100°C	Green	Crimson/blue/crimson
Cold down services	Green	White/blue/white
Hot-water supply	Green	White/crimson/white
Hydraulic power	Green	Salmon pink
Sea, river, untreated	—	Green
Fire extinguishing	Green	Red
Oils		
Diesel fuel	Brown	White
Furnace fuel	—	Brown
Lubricating	Brown	Emerald green
Hydraulic power	Brown	Salmon pink
Transformer	Brown	Crimson
Other suggestions		
Natural gas	Yellow ochre	Yellow
Compressed air	—	Light blue
Vacuum	Light blue	White
Steam	—	Silver grey
Drainage	—	Black
Electrical conduits and ventilation ducts	—	Orange
Acid and alkalis	—	Violet

Standard service codes

Letter symbols are also used to identify pipes and pipelines, fittings, etc. The following summarises British practice.

Water (various)	
Cooling water	CLW
Hot (domestic) water	HWS
Steam	S
Treated water	TW
Wastewater	WW
Boiler-feed water	BFW
Brine	B
Cold water	
Mains	MWS
Down service	CWS
Drinking	DWS
Flushing	FWS
Pressurised	PWS
Cold-water down supply	CWDS
Chilled water	CHW
Fire fighting	
Fire extinguisher	FE
Fire hydrant	FH
Gases	
Town	G
Oxygen	O_2
Nitrous oxide	N_2O
Heating	
Low-pressure water	LPHW
Medium-pressure water	MPHW
High-pressure water	HPHW
Valves	
Air-release	ARV
Air	AV
Auto air	AAV
Ball	BV
Gate	GV
Lockshield	LSV
Non-return	NRV
Pressure-reduction	PRV
Safety	SV
Sluice	SV
Wheel	WV
Sewers	
Foul water	FWS
Surface water	SWS
Drains	
Foul water	FWD
Surface water	SWD
Pipes	
Discharge pipe	DP
Rainwater pipe	RWP
Vent pipe	VP
Effluents	
Foul water	FW
Radio-active water	RAW
Rainwater	RW
Surface water	SW
Fittings	
Bath	b
Bidet	bt
Wash basin	wb
Shower	sh
Urinal	u
Flushing cistern	fc
Sink	s
Drinking fountain	df
Water closet	wc
Manholes, etc.	
Back drop	BD
Invert	INV
Inspection chamber	IC
Manhole	MH
Fresh air inlet	FAI
Position	
High level	HL
Low level	LL
From below	FB
To below	TB
From above	FA
To above	TA
Flow	F
Return	R
Gullies	
Access	AG
Back inlet	BIG
Grease trap	GT
Road	RG
Sealed	SG
Yard	YG
Miscellaneous	
Half-round channel	HRC
Rainwater head	RWH
Condensate	C
Fuel	F
Vacuum	V
Cold feed	CF
Feed and expansion	F & E
Plug cock	PC
Access points	
Access cover	A/C
Cleaning eye	CE
Dry-weather flow	DWF
Fire hydrant	FH
Compressed air	CA
Refrigerants (identified by symbol for particular gas)	R_o
Draw-off point	DO
Open vent	OV
Stop cock	SC

Basic identification colours BS 1710: 1984

Pipe contents	Basic colour names	BS identification colour reference BS 4800
Water	Green	12 D 45
Steam	Silver grey	10 A 03
Oils—mineral, vegetable or animal; combustible liquids	Brown	06 C 39
Gases in either gaseous or liquefied condition (except air)	Yellow ochre	08 C 35
Acids and alkalis	Violet	22 C 37
Air	Light blue	20 E 51
Other liquids	Black	00 E 53
Electrical services and ventilation ducts	Orange	06 E 51
Safety colours		
	Red	04 E 53
	Yellow	08 E 51
	Auxiliary blue	18 E 53
Reference colours (if other than safety colours)		
	Crimson	04 D 45
	Emerald green	14 E 53
	Salmon pink	04 C 33
	Yellow	10 E 53
	Blue	18 E 51

SECTION 2

Valve Types Design and Construction

Plug Valves (Cocks)

Ball Valves

Ball Float Valves

Butterfly Valves

Rotary Disc/Rotor Valves

Globe Valves

Gate Valves

Needle Valves

Pinch Valves

Diaphragm Valves

Slide Valves

Screw-down Valves

Spool Valves

Solenoid Valves

Swing-Check (Flap) Valves

Penstocks

Miscellaneous Valves

Plug Valves (Cocks)

The description 'plug valve' or 'cock valve' is given to the simplest form of valve comprising a body with a tapered or, less frequently, a parallel seating into which a plug fits. The plug is formed with a through-port, the relative position of the port controlling the amount of opening through the valve (Figure 1). A 90° rotation of the plug fully opens or closes the fluid flow.

Greek and Roman periods saw the development of the plug cock valve and it remained virtually unchanged until the 19th century.

The development of the steam engine from the early 18th century led to further valve improvements including the introduction by Timothy Hackworth of adjustable springs instead of weights to the steam safety valve.

The groove-packed plug cock was introduced by Dewarance & Co in 1875, making a valve which was easier to operate and more suitable for steam. In 1886, Joseph Hopkinson introduced the parallel slide valve where the sealing of the valve was produced by line pressure on the disc.

This system is still manufactured today. Plug cock valves are not as efficient as ball valves and can only operate fully open or closed.

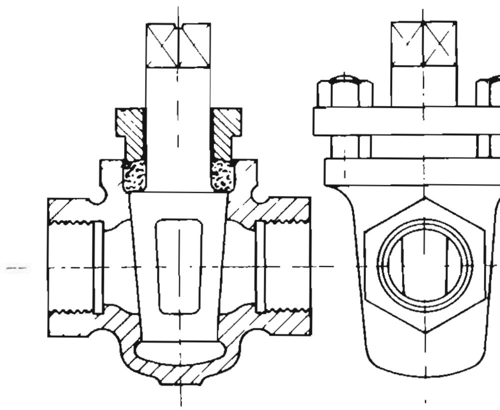
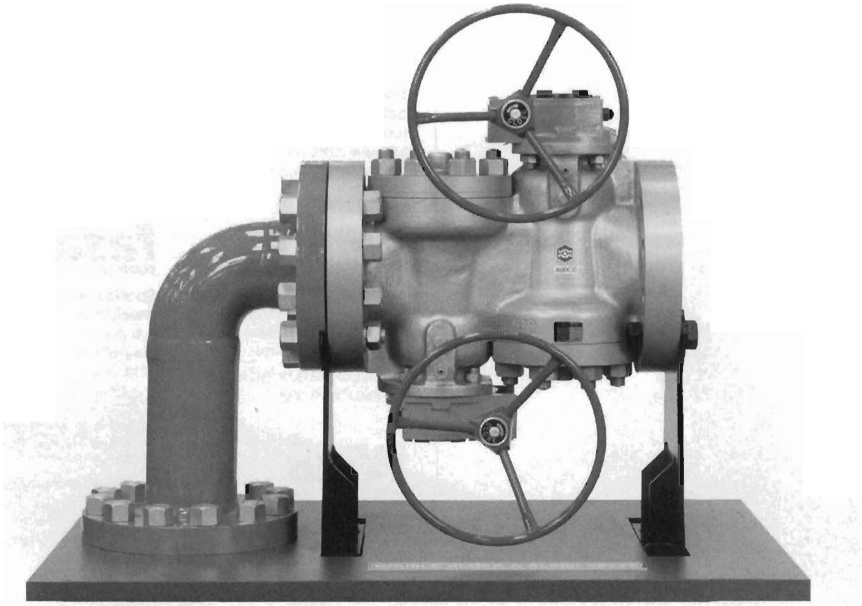


Figure 1.



High-performance pressure-balanced plug valve.

The simple plug valve is generally suitable for low-pressure, low-temperature applications, and can be made in quite large sizes: 250 to 300 mm (10 to 12 in) bore is quite common in some applications. Its main limitation is that if wide variations in fluid temperature are involved, differential expansion is inevitable, leading either to undue stiffness of operation or loss of pressure-tightness.

This can be overcome to some extent by employing a packed gland on which the plug rides (Figure 2). The packing is commonly graphited asbestos. In the smaller range, the sleeve-packed cock represents a distinct step forward in cock design (Figure 3). Not only does this have a perfectly cylindrical plug, more economical to produce than a tapered one, but the resilience afforded by the asbestos fabric sleeve longitudinally compressed by the two plugs screwed

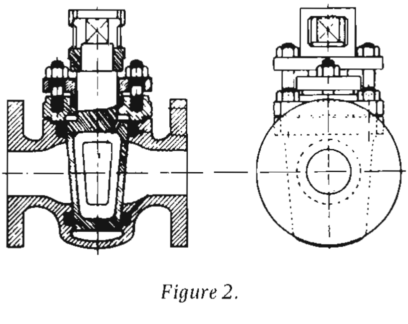


Figure 2.

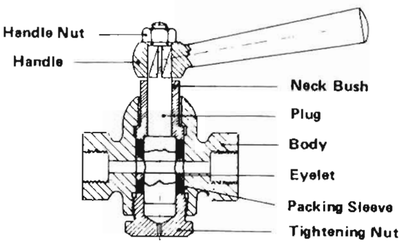
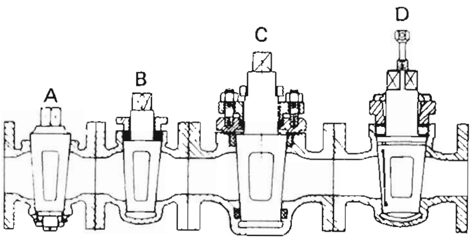


Figure 3.

into both top and bottom of the body provides for temperature variations and thereby prevents binding.

In the UK, the description 'plug valve' is specifically given to a cock which incorporates special design features to reduce the friction between the plug face and the body seat. The plug itself may be tapered or parallel and the movement plain or lubricated (Figure 4). There is also a further variation known as a ball-plug valve, where the plug element is spherical, with circular ports rotating between circular seats of concave section (Figure 5).



A Ground-plug cock with nut and washer base.
B Ground-plug cock gland packed.
C Groove-packed plug cock with gland and holding-down plate.
D Lubricated-plug cock gland packed.

Figure 4.

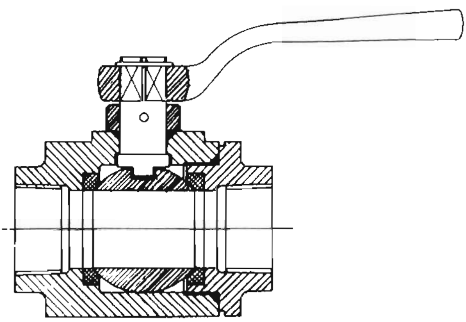
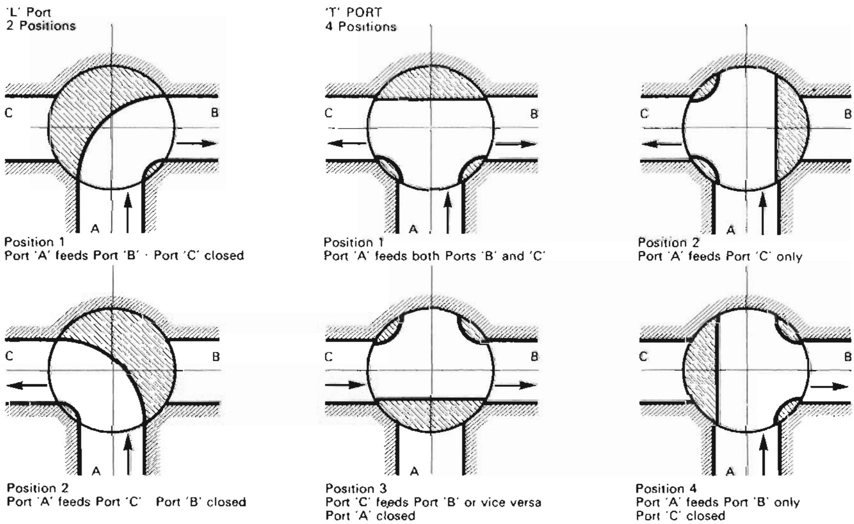


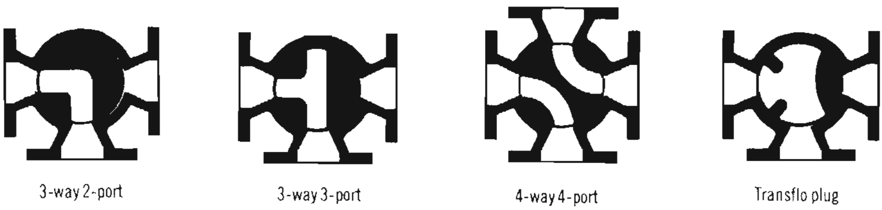
Figure 5.

Plug valves may be further categorised by pattern:

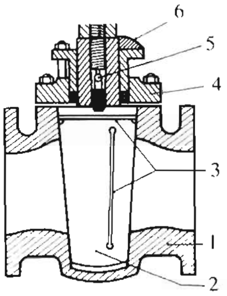
(i) Round opening—with full-bore round ports in both plug and body.
(ii) Rectangular (rectangular opening) with rectangular or similar shaped ports of substantially full-bore section.
(iii) Standard opening—where the area through the valve is less than the area of standard pipe.
(iv) Diamond port—where the opening through the valve is diamond-shaped. Such valves are also normally of venturi design.
(v) Multi-port—with three or more pipe connections, used mainly for transfer or diverting services.
(vi) Venturi design—with reduced-area porting (down to 40%) and featuring venturi flow through the body.
(vii) Short—with reduced-area ports and/or reduced face-to-face dimensions.
(viii) Vertical—with reduced-area seating ports and the plug passages reduced in section to form a throat.



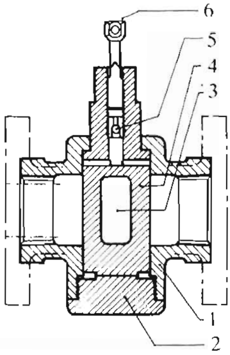
Operation of three-way cocks with 'L' and 'T' ports.



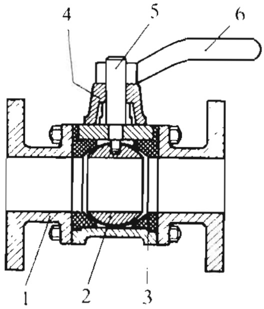
Examples of multi-port arrangements.



Taper-plug valve (lubricated)
1. Body
2. Plug
3. Lubricant grains
4. Cover
5. Lubricant check valve
6. Gland follower



Parallel-plug valve
1. Body
2. Bottom cover
3. Plug port
4. Plug
5. Lubricant grains
6. Lubricant screw



Ball-plug valve
1. Body
2. Ball
3. Seal
4. Bonnet
5. Spundle
6. Handle

Materials

Cocks and plug valves are produced in a variety of metals and plastics and also include lined types. Metals most commonly used are brass, bronze, steel and stainless steel.

Basic design proportions

A rectangular- or trapezoid-section port is commonly preferred as this can be accommodated in a plug of smaller diameter than that required for a circular port of the same area. The width of the port is then often made less than half of the bore to provide an effective positive lap for sealing. The length of port is then given by d/2, where d is the pipe diameter. In practice a small addition is usually made to this length to allow for radiusing the corners of the opening.

In the case of multi-port cocks or plug valves, negative lap may be called for to ensure that there is no complete shut-off during the transition of ports. This applies particularly when connected to a positive displacement pump (i.e. to prevent the pump pumping against a closed outlet).

See also the chapter on Ball Valves.

Pressure-balanced taper-plug valves

In larger taper-plug valves, pressure-balanced plugs are fitted for pressure pulsing or very high static pressure applications. With a non-pressure

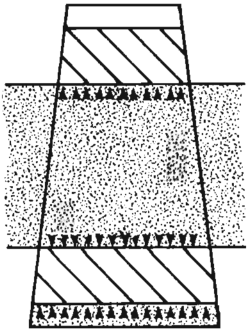


Figure 6(a). Non-pressure balanced taper plug.

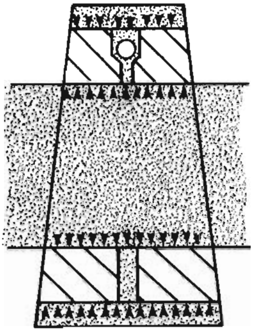


Figure 6(b). Pressure-balanced taper plug.

balanced plug, line pressure in an open valve can find its way into the large end chamber which exists below the plug. Under these conditions a resultant force exists tending to push the plug into its tapered seat with the danger of taper locking causing a seized valve, as shown in Figure 6(a). This resultant force persists whether the line pressure subsequently remains high or is reduced.

The development of an out-of-balance force on the plug is not an inevitable event with ordinary taper-plug valves, as there is normally sealant pressure acting on the small end of the plug. Nevertheless it can occur and can cause valve seizure.

With a pressure-balanced valve, the live-line pressure is used to replace sealant pressure by allowing the line to pressurize the small end chamber. A balancing force is produced which prevents taper lock without the need for sealant pressure. Figure 6(b) shows how a more balanced position is reached when line pressure is allowed to equalise the pressure acting on the end of the plug.

The pressure-balance system consists of two holes in the plug connecting chambers at each end of the plug with the line pressure. The hole in the small end of the plug contains a non-return valve. This enables sealant pressure to be built up if necessary, while allowing access of the line pressure to the small end chamber. Thus the pressure in the large end chamber always equals line pressure and the pressure in the small end chamber is always equal to, or greater than, the line pressure.

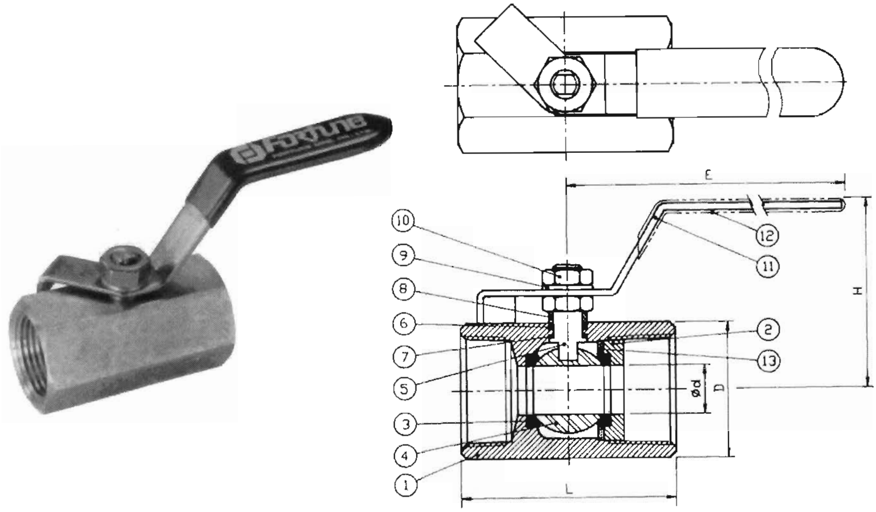
Ball Valves

The ball valve, or spherical-plug valve as it is sometimes known, was developed around 1936, although the idea of a ball valve dates back to ancient times. Modern ball valves, depending on type and pressure class, should be designed in conformity with international standards, e.g. BS 5351, API 6D and ANSIB 16.34. Normally, ball valves have polymer-based seals.

Ball valves are among the least expensive but most widely used of all valve types, as well as being available in an extremely wide range of sizes. Basic geometry involves a spherical ball located by two resilient sealing rings in a simple body form (Figure 1).

The ball has a hole through one axis, connecting inlet to outlet with full-bore flow when aligned with the axis of the valve. Rotating the ball through 90°

A typical range of ball valves.



Materials list:

No.	Part	Specification	Quantity
1	Body	ASTM A351-CF8M(316)	1
2	Inscrewed seat retainer	ASTM A351-CF8M(316)	1
3	Seat	R. TFE	2
4	Ball	ASTM A351-CF8M(316)	1
5	Stem	ASTM A276-316	1
6	Stem packing	PTFE	1
7	Thrust washer	PTFE	1
8	Gland	AISI 304	1
9	Stem washer	AISI 304	1
10	Handle nut	AISI 304	1
11	Handle	AISI 304	1
12	Handle cover	Plastic	1
13	Gasket	PTFE	1

Figure 1. Standard ball valve.

completely closes the flow passage with positive sealing via the sealing rings. Sealing is equally effective in both directions.

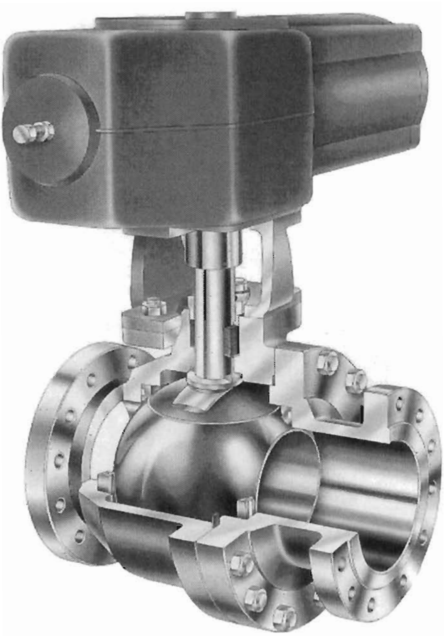
Body forms and matching ball hole may provide straight-through (full-bore parallel), reduced flow, or venturi flow. Ends can be flanged or threaded.

The ball itself may be free floating, in which case the squared off or splined end of the stem fits into a matching recess in the top of the ball. On larger valves the ball may be trunnion-mounted. Trunnion mounting reduces operating torque to about two-thirds that of the floating ball (Figure 2).

Ball valves are produced in top-entry and split-body forms for assembly and for renewal of the seals and ball. They are also produced in multi-port configurations, thus normally requiring a larger size of ball to accommodate multi-port drillings. These ports can be proportioned to give positive lap or negative lap as required (see also Figure 3).

Cut-away section of a multi-port valve.

Section view clearly illustrating how a characterised 'V' seat allows for precise flow control in a modulating ball valve.



Flanged full-bore ball valve.

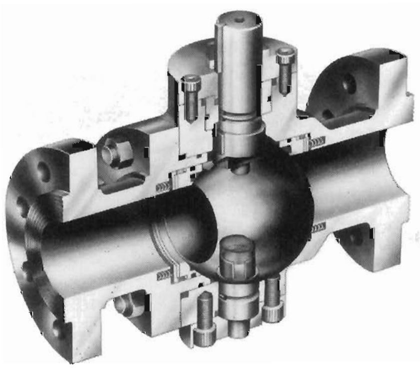


Figure 2. Trunnion-mounted ball valve.

Full operating movement is 90° rotation of the ball. Steps may be incorporated to limit movement of the operating lever, or continuous rotation may be possible. In either case the lever position is in line with the axis of the valve in the open position and at right angles to it in the closed position. Larger ball valves may be operated by handwheels through reduction gearing, or by powered actuators. In all cases opening/closing torque is low because the only friction forces involved are those of the ball rotating against its seals and the friction offered by the stem gland. The latter can range from O-rings to glands

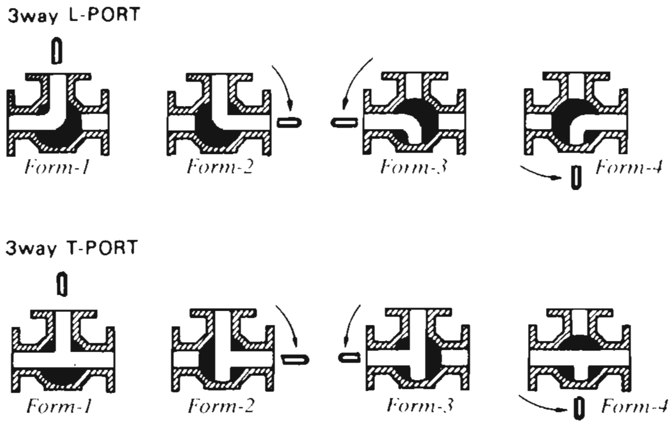
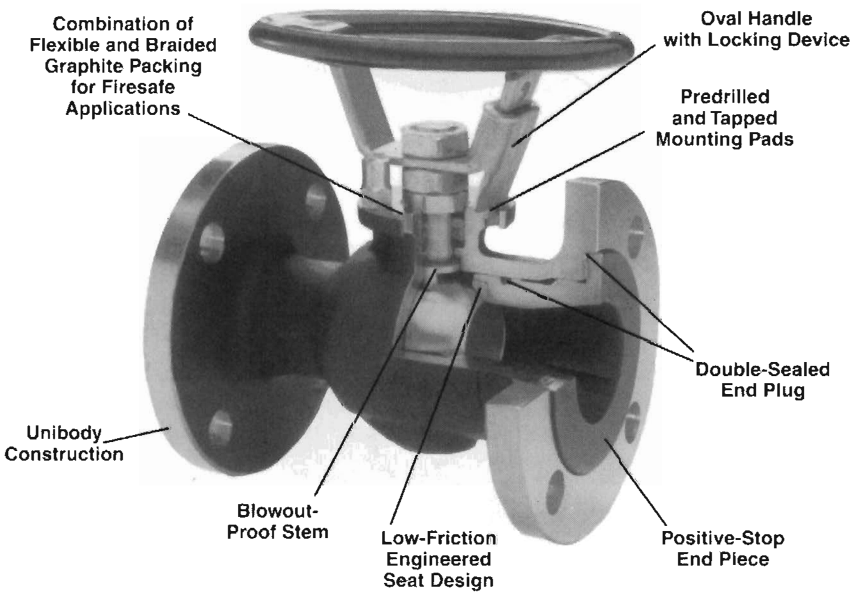
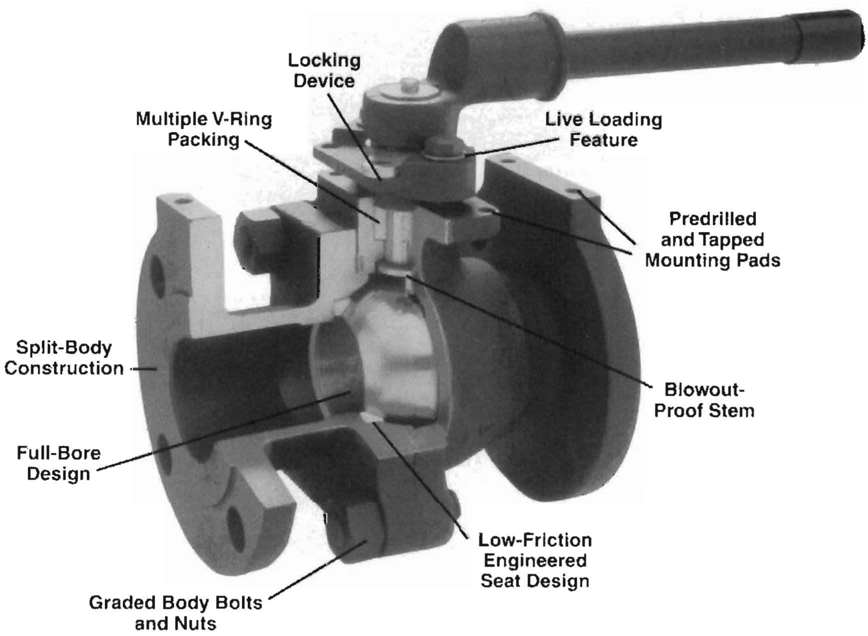


Figure 3. Three-way ball valve.

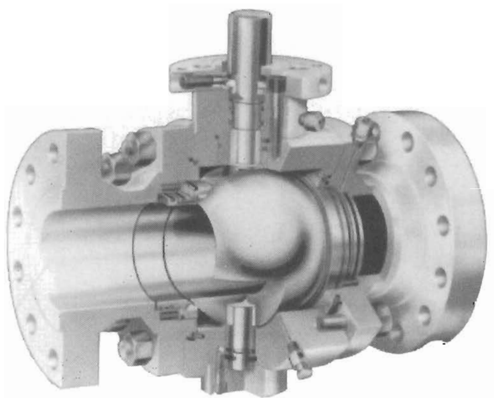


One-piece (unibody) ball valve.

fitted with die-formed packing rings. In some ball valves the ball is held against the seat by the cam action of a specially shaped stem. By turning the valve handwheel the ball is pulled away from the seat before being rotated. A precision spiral groove turns the stem and ball 90°, without ball-to-seat friction, to full-straight through-flow when open. The reverse action lowers the stem, turning the ball to the closed position, and the final handwheel turn tilts the ball and mechanically wedges it against the seat to seal the valve closed.



Two-piece (split body) flanged ball valve.



Full-port high-pressure ball valve.

Historically, ball valves have been produced with soft, not metal seats because generally soft seats have covered most applications satisfactorily.

Many valves of this type have seals made from PTFE, compounded with graphite, glass or steel powder to improve the material properties. However, abrasive media, high pressures and high temperatures can severely stress the polymeric seals normally used and lead to damage (Figure 4).

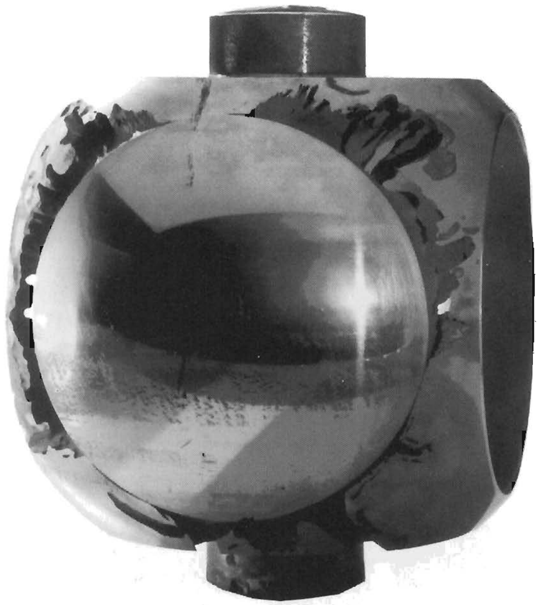


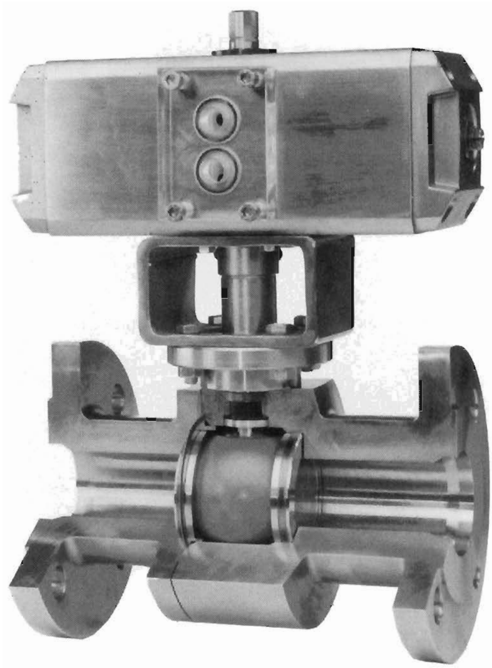
Figure 4. This ball was taken from a valve that had seen 3 years service in a cement works. The polymer sealing rings have been destroyed, the ball and the body severely damaged. Metallic seals can prevent such problems.

For nominal diameters of # DN50 PTFE, seals can only be loaded to a full pressure of PN100 up to a temperature of approximately 100°C; with nominal diameters above DN80, the operating pressure is limited to 50 bar. Only gradual improvements can be made if highly resistant polymers such as POM are used. Upper temperature limits are 250°C with huge restrictions on pressure/load capacity.

Metal-seated ball valves

Metal-seated ball valves first came to prominence in the 1960s. They offer a number of advantages including: tight shut-off, smooth control, no jamming, low torque, wide temperature range, good corrosion and wear resistance and stability under pressure. The greatest risk to metal-seated ball valves is posed by corrosion through pitting, fretting corrosion, intercrystalline corrosion and stress corrosion cracking. Media that contain even low quantities of aggressive substances are capable of causing corrosion.

Metal seals do not bed in as easily as soft seals under pressure. It is therefore important for the ball and sealing rings to be machined precisely and have both hard- and low-friction coatings applied to the base material.



Ball valves with metallic seals are suitable for use in high-solids abrasive media, for high and low temperatures, for extreme operating pressures, and for frequent operation. Even with critical media, they can be used for flow regulation. This pneumatically-activated ball valve is an ideal component for increasing plant safety. In the event of failure of the compressed-air supply the spring-loaded, pneumatic drive closes the valve automatically, rapidly and reliably.

Metallic seats tend to employ both nickel- and cobalt-based alloys and elements such as chrome and tungsten. However, the trend appears to be towards the use of different surface coatings for ball and sealing rings and choosing between them to suit the various circumstances.

With the seat-supported ball valve (Figure 5), the valve seals on the downstream side. The upstream pressure pushes the ball against the downstream seat, closing it tightly.

In the trunnion-mounted ball valve with a bellows seat (Figure 6), the valve seals on the upstream side. The internal pressure expands the bellows axially, pushing the seat against the ball. The seat is pressure-assisted and spring-energised. The bellows seat acts as the seating component. This type of ball valve is suitable for the most demanding on–off services.

The special control seat shown in Figure 7 works like a normal pressure-assisted seat in trunnion-mounted ball valves. The upstream pressure is led through the hole behind the seat, pushing it against the ball.

The seat is spring-energised to ensure low-pressure tightness. In control, the high-velocity flow passes through the restriction point of the partly open

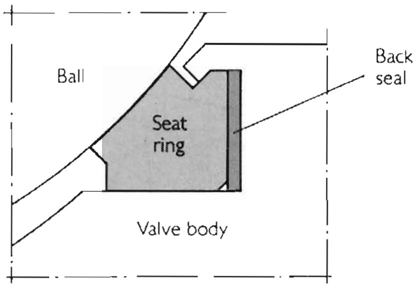


Figure 5. The metal seating principle in a typical seat-supported ball valve.

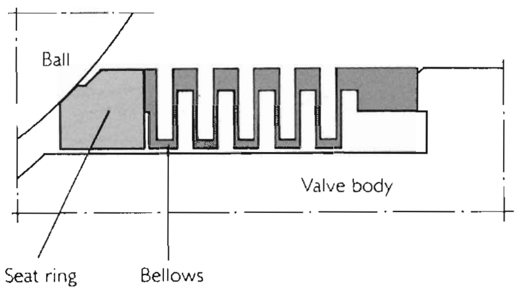


Figure 6. The bellows seat for a typical trunnion-mounted ball valve.

valve. The high velocity creates low pressure, which is led behind the ball seat through the hole located in the vena contracta. The seat will thus be unloaded.

The sealing principle of the floating-ball valve example shown in Figure 8 is effected at the downstream seat where the ball is pressed against the opposite seat by the medium pressure. In doing so the seat rings have a double function. They seal off and at the same time serve as a bearing. The seal at the upstream seat can be relieved in order to avoid a build-up of pressure.

The sealing principle of the fixed-ball valve example shown in Figure 9 is one where the sealing is effected at the upstream seat where the spring-supported seat is pressed against the fixed ball by the medium pressure.

The ball itself can be fixed by bearing pads in the body, by trunnions or by bearing stems. A pressure build-up is prevented by the spring-supported seats in connection with the fixed ball. To summarise, effective sealing depends on:

- the contact pressure
- the contact surface of the seat
- the accuracy of the surface finish on the ball and ball seat
- the sealing design and the sealing material

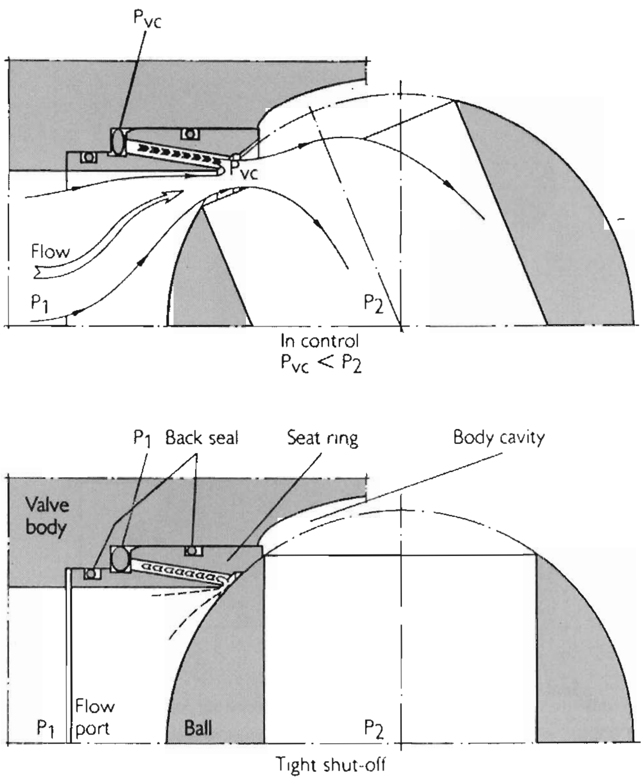


Figure 7. The special control seat.

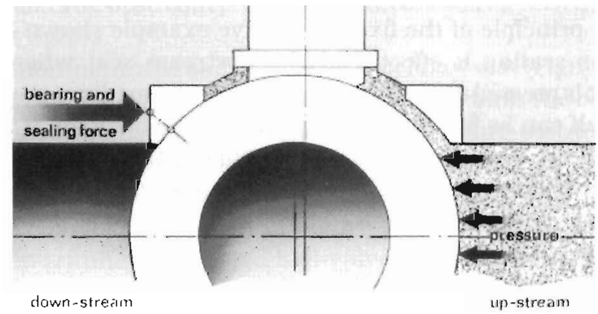


Figure 8. The sealing principle of the floating ball.

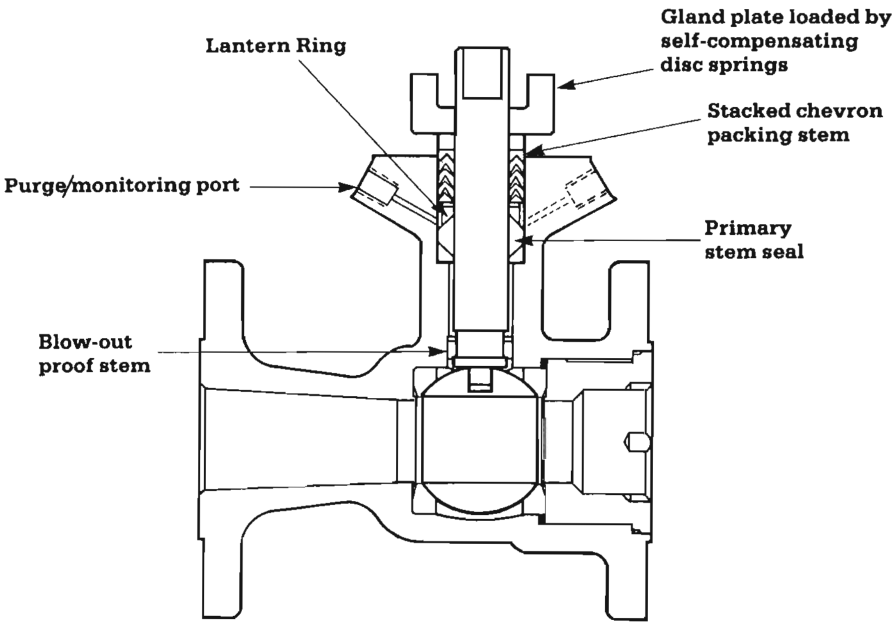


Diagram showing the dual-stem sealing arrangement in a high-integrity ball valve.

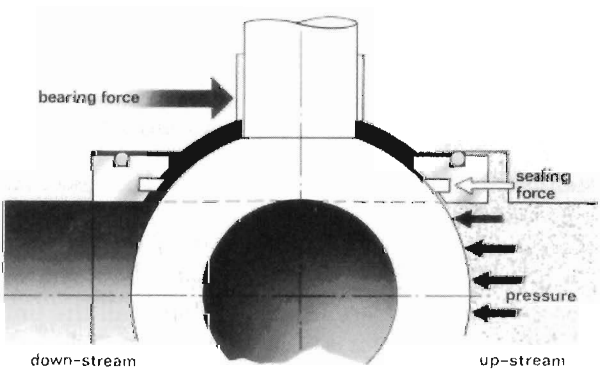


Figure 9. The sealing principle of the fixed ball.

Generally, ball valves are sealed by applying a load to a soft seating material between the valve body and ball to create localised yielding.

Seals of plastic material usually depend on localised yield to achieve bubble-tight sealing. The problem with a jam seat is that increasing the shut-off pressure can increase plastic deformation. As long as pressure remains at a high level this is not a problem; leakage may occur if the shut-off pressure is decreased. A jam seat has no pressure compensation. Another area to consider about jam seats is temperature swings.

With increasing temperatures, metallic ball and valve casings expand. PTFE valve seats expand at a much higher rate and if the temperature change is high enough, the jam seat will tend to generate a 'self stress' above its yield strength and deform plastically beyond its initial state. When the valve is cooled, shrinkage of the additionally deformed seat may result in leakage.

A possible way of overcoming this is to employ valves with flexible lip seats (Figure 10) or seats that incorporate a separate double block and bleed design. Another aspect to consider with soft-seated ball valves is built-in body-cavity pressure relief. Ordinary water trapped in a valve cavity without air will increase in pressure by about 100 lb/in^2. The pressure/temperature relationships of most common liquids are in the order of 90–110 lb/in^2 per °F (11.2—13.7 bar per °C).

The cavity area created by the two soft seats of a ball valve is a typical area for pressure increases. While the valve is open, any pressure in the cavity zone created by the ball, seat and body can be vented via a hole from the bottom of the stem slot to the ball waterway.

In the closed position, relieving cavity pressure is more difficult. Some valves have a vent hole in the ball.

Cavity-pressure increases derive from the differential thermal expansion rates of incompressible fluids and typically a venting of one hundredth of a cc of trapped liquid will bring cavity pressure back to normal.

The key to ball-valve performance is the sealing (seating) structure, regardless of whether the seats are metal or plastic.

Pressure–temperature ratings

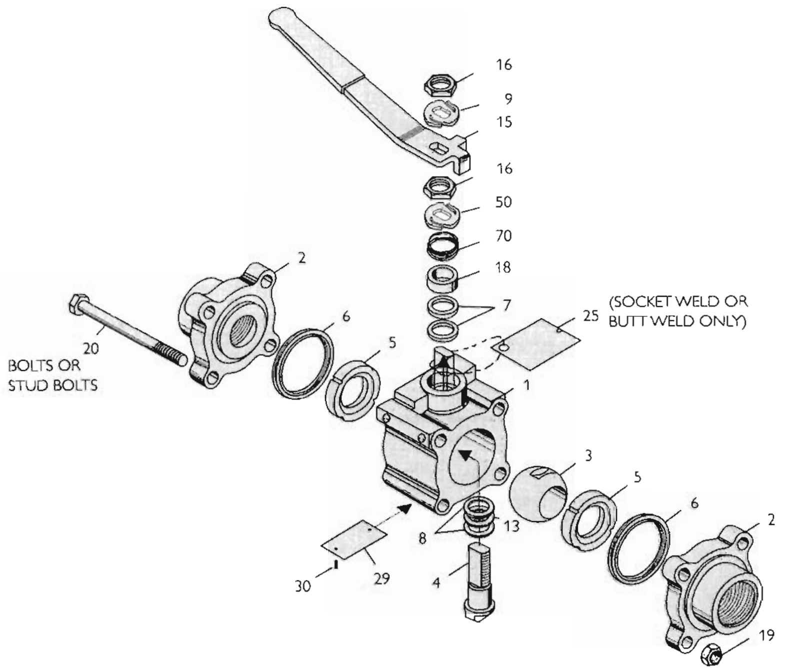
The pressure–temperature ratings of soft-seated ball valves are determined not only by the valve body materials, but also by the sealing material used for ball seats. Sealing materials for seats may be PTFE, 15% or 25% glass-filled PTFE, FPM, Celastic, N.R.G., POM, Lyton and steel.

New and improved polymers are being developed all the time for coating and sealing and the use of ceramics is becoming widespread.

It is very difficult to pre-determine exact pressure–temperature ratings for all kinds of media under all imaginable conditions. The chart shown in Figure 11 gives a typical general overview.

Pressure–temperature seat ratings indicated by the solid lines on the chart are based on differential pressure with the ball in fully-closed position and refer only to seats. The dotted lines indicate the maximum working pressures for carbon-steel valve bodies made from TstE 355N (equal to ASTM A350 LF2). For ratings of other body materials refer to ANSI B 16.34.

Pressure–temperature seat ratings for metal-seated valves are the same as the body ratings.



Standard parts list

Item no.	Part name	Body material: Carbon steel	Body material: Stainless steel
1	Body	ASTM A105N, ASTM A216 WCB (0.25% C Max) or BS970 070 M20	ASTM A812 F 316, ASTM A351 CF8M or BS970 S316
2	Body cap	ASTM A105N, ASTM A216 WCB (0.25% C Mac) or BS970 070 M20	ASTM A812 F 316L ASTM A351 CF8M/CF3M or BS970 S316
3	Ball	316 stainless steel	
4	Stem	316 stainless steel or 17-4 PH stainless steel	
5	Seat	PTFE (T), filled PTFE (M) or acetal resin (R) (Delrin*)	
6	Body seal	Spiral-wound 316 stainless steel and graphite	
7	Gland packing	PTFE	
8	Stem-thrust bearing	Carbon-filled PTFE or acetal resin	
9	Stem-tab washer	Stainless steel	
13	Stem-thrust seal	Graphite	
15	Handle	Stainless steel with PVC sleeve	
16	Stem nut	316 stainless steel	
18	Gland	316 stainless steel	
19	Nut	ASTM A194 grade 2H or ASTM A194 grade 2HM, 7M	ASTM A194 grade 8B, 8Cb, 8TB or ASTM A453 grade 660
20	Body bolt/stud	ASTM A193 grade B7 or ASTM A193 grade B7M, L7M	ASTM A193 grade B8, B8C, B8T class 2 or ASTM A453 grade 660
25	Weld warning tag	—	
29	Identification plate	Stainless steel	
30	Drive pin	Stainless steel	
50	Stem-tab washer	Stainless steel	
70	Anti-static spring	Stainless steel	

Figure 10. Soft-seated ball valve, exploded view.

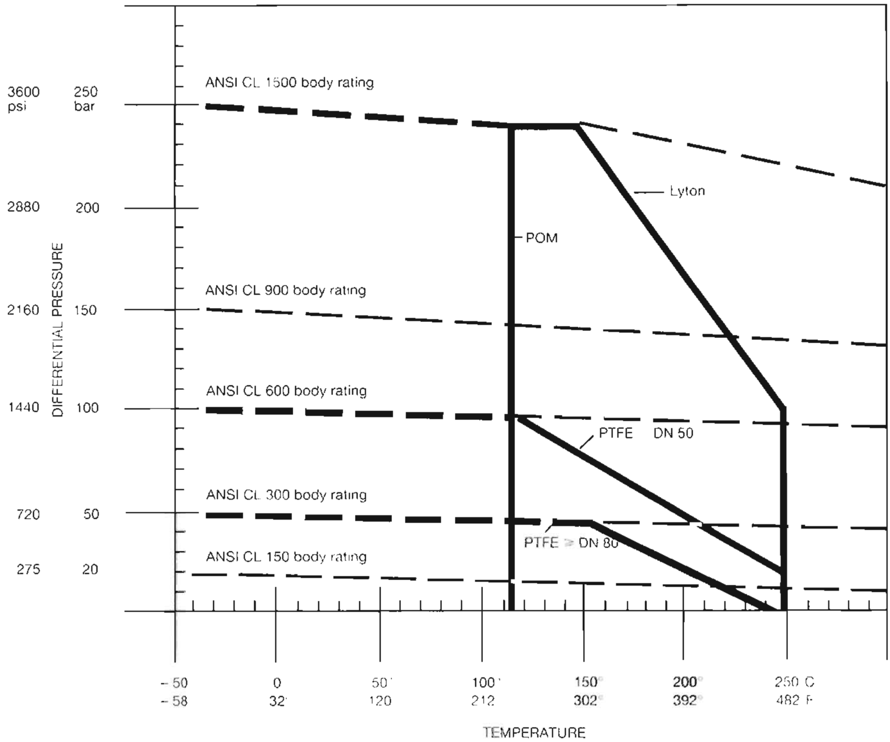


Figure 11. Pressure–temperature ratings.

Flow data

Typical flow data is shown in Table 1. The flow rates were determined for ball valves in fully-open position and a water temperature of 15°C (59°F).

Typical major application areas for ball valves include:

Refineries

- shut-off and isolation valves for tower bottom lines and thermal-cracking units with coke problems
- gas/oil separation
- gas distribution including measuring, metering and pressure regulation stations
- controlling oil loading
- pumping and compressor stations
- emergency shut-down
- refining units

Table 1. Flow data

Full bore
Nominal flow rate

Nominal size mm	Nominal size in	Kv m^3/h	Cv US gallons per min
15×15×15	1/2×1/2×1/2	19.4	22.6
20×20×20	3/4×3/4×3/4	45.6	53.2
25×25×25	1×1×1	71.5	83.4
40×40×40	1 1/2×1 1/2×1 1/2	170	198
50×50×50	2×2×2	275	321
80×80×80	3×3×3	905	1056
100×100×100	4×4×4	1414	1650
150×150×150	6×6×6	3674	4288
200×200×200	8×8×8	7155	8350
250×250×250	10×10×10	12,500	14,590
300×300×300	12×12×12	20,780	24,250
400×400×400	16×16×16	37,000	43,100
500×500×500	20×20×20	70,700	82,500

Reduced bore
Nominal flow rate

Nominal size mm	Nominal size in	Kv m^3/h	Cv US gallons per min
—	—	—	—
20×15×20	3/4×1/2×3/4	14.3	16.7
25×20×25	1×3/4×1	40.1	46.8
40×32×40	1 1/2×1 1/4×1 1/2	89.8	105
50×40×50	2×1 1/2×2	146	170
80×65×80	3×2 1/2×3	484	564
100×80×100	4×3×4	800	934
150×100×150	6×4×6	728	850
200×150×200	8×6×8	3577	4175
250×200×250	10×8×10	6933	8090
300×250×300	12×10×12	11,392	13,294
400×300×400	16×12×16	1600	18,672
500×400×500	20×16×20	33,333	38,900

Kv value is the full-capacity flow rate through the ball valve in cubic metres per hour (m^3/h) with a pressure drop of 1 bar.
Cv value is the full-capacity flow rate through the ball valve in US gallons/min of water at 60°F with a pressure drop of 1 psi.

Chemical and petrochemical complexes

- low differential pressure control
- emission control
- handling highly viscous fluids, abrasive slurries or corrosives as well as non-corrosives in processes and storing facilities

Power industry

- boiler feedwater control
- control and shut-off for steam
- burner trip valves
- sluicing valves for feeding coal into pressurised combustors and for extracting fly ash

Gas and oil production

- subsea isolation and shut-down
- well-head isolation
- pipeline surge control
- secondary and enhanced oil recovery
- processing separation
- transmission and distribution
- storage

Pulp and paper industry

- pulp mill digesters
- shut-off valves
- batch-digester blow service
- liquor fill and circulation
- lime mud (slurry) flow control
- dilution water control

Other common areas for the application of ball valves include: food industry, water supply and transport, marine and solids transport.

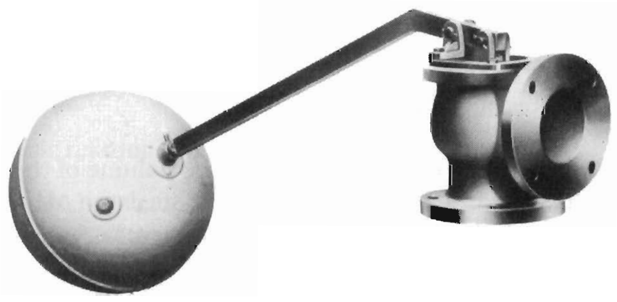
Ball Float Valves

The typical ball float valve consists of a control valve of the piston and disc-type operated by a floating ball and lever mechanism adjusted to open the valve at a predetermined liquid level. It is thus essentially a level-control valve and is used mainly for controlling the supply of make-up water (e.g. in cisterns, water tanks, etc.)

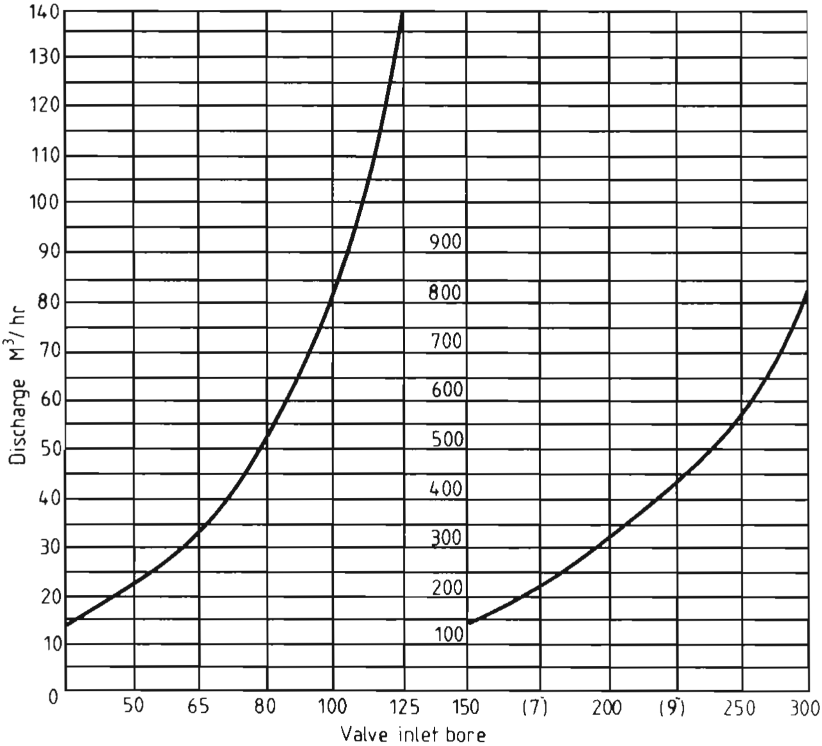
The most effective type is the equilibrium ball float valve (Figure 1), so called because the upward and downward pressure forces are nearly balanced out (leaving just enough unbalance to eliminate hunting). The basic type finds widespread application. The body is usually of angle pattern with the inlet flanged and bolted to the inlet pipe flange with the tank wall sandwiched between.

Design geometry calculation

Design geometry calculations for ball float valves can be tedious. As far as fluid forces are concerned there is an upward force at the valve position due to mains pressure tending to force the float downwards, which will normally be resisted by the displacement force generated by the float under equilibrium conditions (Figure 2). These equilibrium conditions correspond to the valve being held closed by a surplus of 'displacement' force. (In the event of the water level falling, of course, the valve will open to allow inflow of water until equilibrium conditions are restored.)



A typical ball float valve.



Headloss approximately 1.5 m angular and 4.5 m straight through.

Typical ball float valve capacity chart.

Specifically, the valve force (V_F) effective at the ball is given by:

$$V_F = \frac{0.7854\ PN}{L} \times d_2^{\,2} - d_1^{\,2}$$

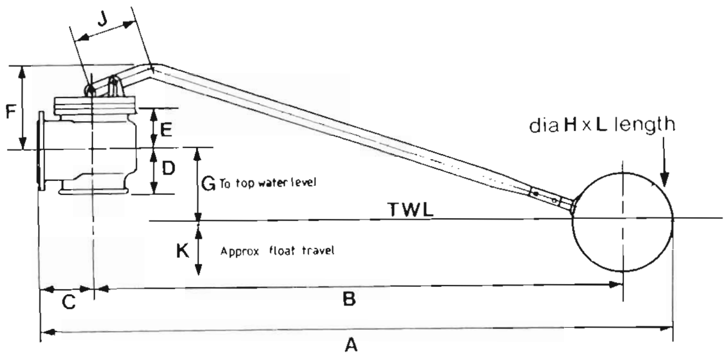
where

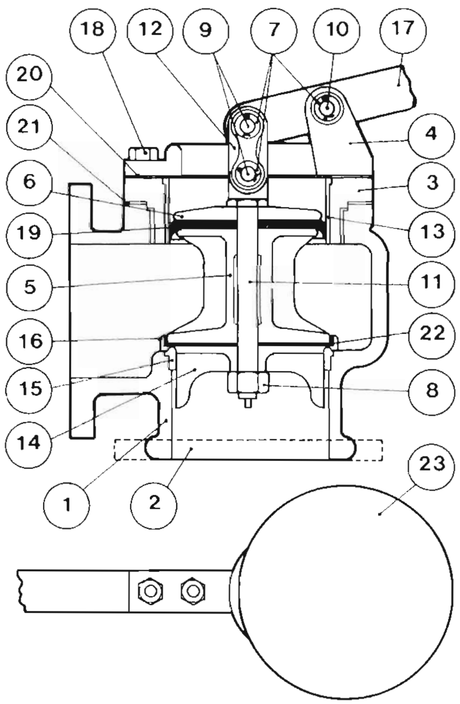
P = supply water pressure
N = fulcrum, distance from valve
L = fulcrum, distance from ball float
d_1 = outside diameter of valve-sealing face
d_2 = diameter of valve piston.

The displacement force (D_F) is governed by the volume of the ball float (or diameter in the usual case of a circular float) and its depth of immersion. It is usual in the case of spherical-ball floats to design for 50% immersion, when:

$$D_F = 0.01\ D^3 \text{ (to a close approximation)}$$

where D is the float diameter.





Item description

1 Body (beaded outlet)
2 Body (flanged outlet)
3 Cover
4 Fulcrum and retaining ring
5 Valve
6 Keep plate
7 Split pins and washers
8 Nut
9 Pins to links
10 Pins to fulcrum
11 Spindle
12 Links
13 Liner
14 Winged-valve plate
15 Seat ring
16 Keep ring
17 Lever
18 Studs and nuts
19 Cup washers
20 Cover joint
21 Body joint
22 Seat washer
23 Float

Figure 1. Angular-pattern ball float valve.

Thus the basic relationship is modified by:

(i) The necessity of maintaining a positive closing force on the valve for fluid tightness. An empirical figure here is to make D_F equal to $1.25 \times V_F$.

(ii) Additional movements introduced by the weight of the lever on both sides of the fulcrum or pivot point and the weight of the float. These can be calculated or eliminated by counterbalancing.

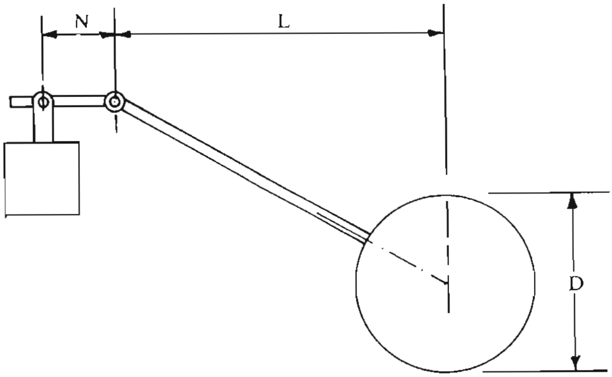


Figure 2. Ball float valve displacement force generated by the float.

(iii) Frictional resistance of the pivot and rising part(s) of the valve. This will call for an additional increase in displacement force, i.e. float size. It is difficult to establish required values except on empirical lines as frictional forces may be subject to change with age.

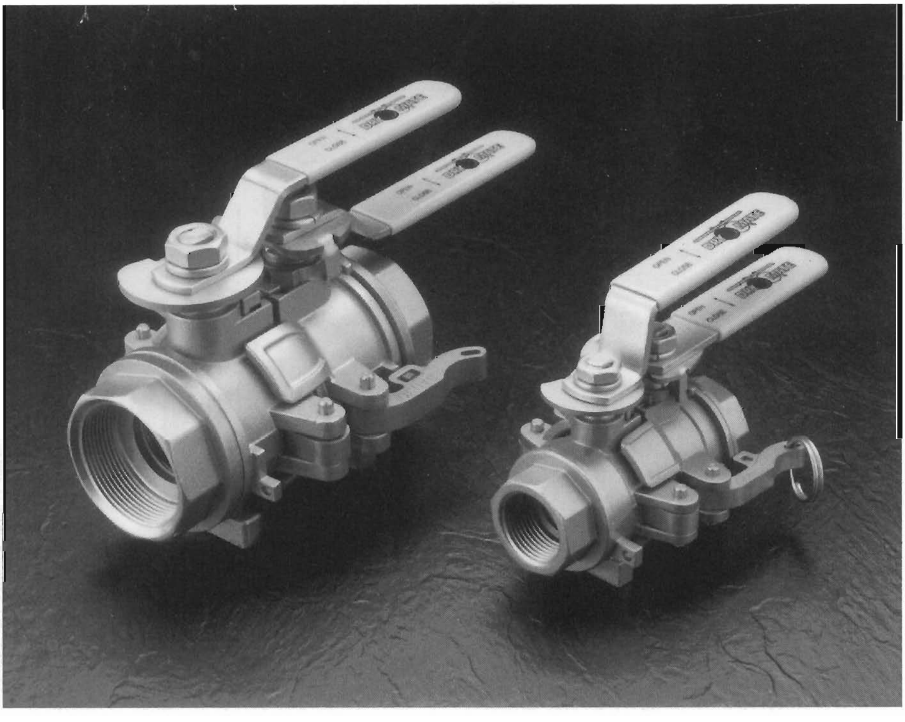
It may be further necessary to evaluate the displacement force with the float fully submerged (when $D_F = 0.02\ D^3$) and the resulting loading on the fulcrum pin. Design should allow adequate sheer strength on the pin to allow for this contingency, particularly in ball float systems used with level controls in large tanks and/or inaccessible positions.

Butterfly Valves

Since it was first introduced well over one hundred years ago, the butterfly valve has come through many development stages to become one of the most successful high-performance valves in use today.

The first butterfly valve was used by James Watt, the Scottish engineer, for his steam engine. It was also used in the Mercedes motor car in 1901 for the fuel intake linked to the accelerator pedal.

Initially, butterfly valves were limited to low pressure drop applications and sizes that were about six inches or more. They also tended not to seal too well.



Butterfly valves for industrial applications.

In basic terms, a butterfly valve uses a flat disc in which the closure device rotates about an axis regulating the flow of liquids with off-centre or in-line sealing. Smaller sizes usually have manual operation, but much larger devices such as penstock control valves are usually motorised.

Butterfly valves can offer attractive cost savings and operating benefits over conventional globe valves. With high flow capacities, butterfly valves enable the use of smaller units which reduces cost, weight and space requirements. With only two wetted parts and a range of valve linings, butterfly valves isolate the body from the media, thus eliminating the need for expensive exotic materials.

The smooth contoured, crevice-free disc produces lower torques, while the butterfly valve design makes it easy to install, maintain and service.

The two main groupings are general purpose butterfly valves and high-performance butterfly valves. Butterfly valves have the disadvantage that they restrict the flow through the pipe and solids catching on the disc can cause a blockage or prevent the valve from closing. Butterfly valves are used in high-temperature, high-pressure applications or those involving toxic or corrosive fluids. Modern butterfly valves are normally of wafer design, fitting

Butterfly valve for the chemical industry.

directly between pipeline flanges. Double-eccentric (offset) design butterfly valves have the sealing plane of the disc offset from the axis of rotation to create an interrupted sealing surface and the axis of the disc is laterally displaced from the true centre of the disc so that the disc will 'cam' away from the seat as the valve is cycled open.

Butterfly valve movement is simple and straightforward, requiring only 90° rotation of the butterfly for full movement (or somewhat less in most designs).

The main technologically significant features of a butterfly valve are the designs of the disc and the seat. Nearly all modern high-performance butterfly valves incorporate some form of double-offset design as previously described.

The two offsets in a typical butterfly disc are shown in Figure 1.

Body construction is normally cast iron, ductile iron, cast steel, bronze and epoxy. Various other materials may be used depending on size and application. Welded materials (e.g. steel, stainless steel and titanium) may be used for certain valves and in particular where percolation of gases through cast components is to be prevented.

Discs are also usually of cast iron, although again alternative materials may be specified for particular services. Profile shapes vary, most having some form of convex streamline shape to minimise head loss.

Valve seats

The seat (closure member) of the butterfly valve is an area where there are as many designs as there are manufacturers of butterfly valves. A single-piece

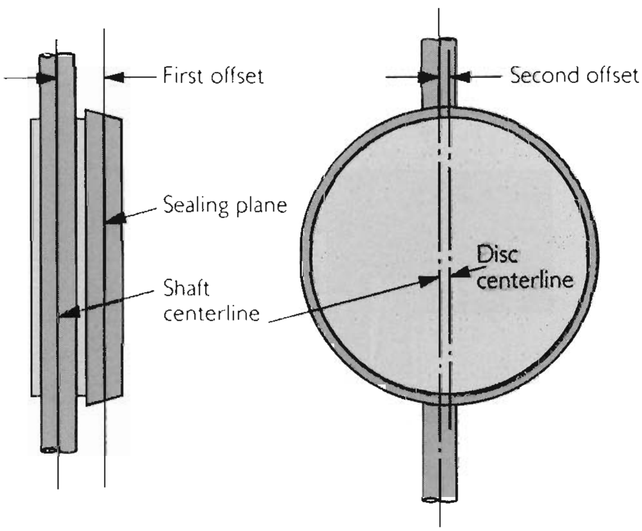


Figure 1. The two offsets in a butterfly disc.

Butterfly valve for the water industry.

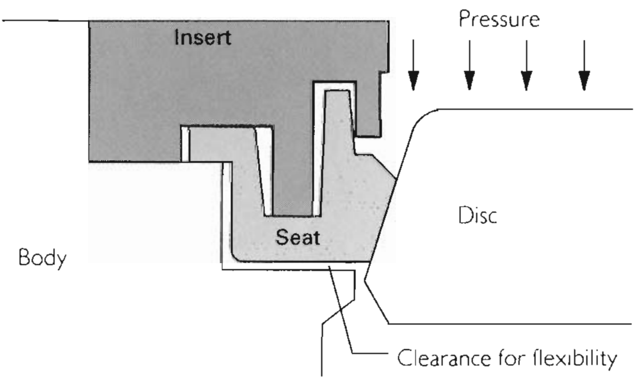


Figure 2. Wafer-sphere sealing principle with the disc downstream of the seat.

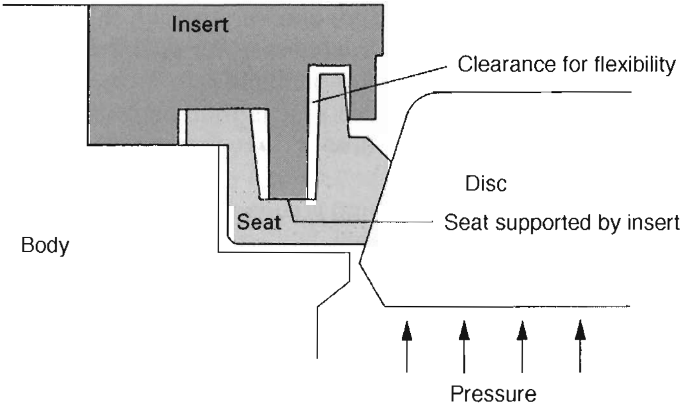
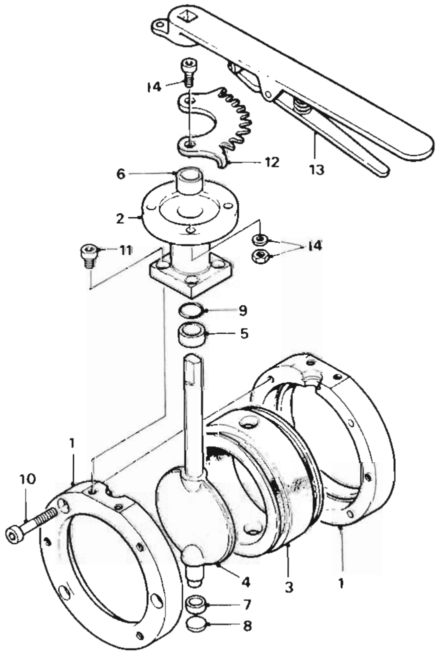


Figure 3. Wafer-sphere sealing principle with the seat downstream of the disc.



1	Body cartridges	8	Plus
2	Top adaptor	9	'O-ring'
3	Seal	10	Body securing screws
4	Disc/Stem	11	Top adaptor securing screws
5	Centre bush	12	Notch plate
6	Top bush	13	Lever
7	Bottom bush	14	Notch plate fixings

Figure 4. A wafer-type butterfly valve.

flexible PTFE seat has proved to be a popular choice, with thick cross-sections throughout the seat, pre-compression of the seat for low-pressure sealing and clearances surrounding the seat to allow flexibility.

The significance of the single-piece seat is that there are no O-rings or metallic springs to limit the temperature or corrosive conditions that the PTFE seat can be exposed to.

A typical example of PTFE seat design is shown in Figure 2 where the disc is downstream of the seat.

As line pressure is applied, the full cross-section of the seat is pressurised, which causes the seat to follow the natural deflections of the disc under pressure. Pressure activation of the seat enhances sealing with increasing line pressure, despite the fact that the disc is moving away from the seat due to the same pressure.

With the flow in the opposite direction, with the seat downstream of the disc, the seat is supported by the seat retainer (Figure 3). The disc is deflected by pressure into the seat, enhancing the sealing as the pressure increases.

A wafer-type butterfly valve is shown in Figure 4. Typically the valve consists of a stainless steel disc/stem in a polymer seat, fastened by two half-body cartridges. This particular valve has three side thrust-absorbing bushes, one top and bottom and one directly above the seat with a secondary O-ring seat adjacent.

Elastomer-lined centred-disc butterfly valve for water supply.

Fully rubber-lined butterfly valves, similar to the type shown in Figure 5, incorporate a rubber lining that is bonded and integral with the valve body. The main advantage here is that there is little or no deformation of the lining or corrosion to the body.

Ebonite-lined butterfly valves are designed for use in systems which carry seawater and other moderately aggressive liquids. Ebonite is a hard natural or synthetic rubber.

Ceramic- and fluoropolymer-lined valves are gaining in popularity for controlling highly corrosive and abrasive liquids, gases, slurries and powders.

Another basic arrangement is a corrosion-resistant seat (e.g. bronze or stainless steel), into which a continuous rubber-ring seal is fitted. Others include flexible metal-seal rings (for high-temperature services) (Figure 6). Detail design may provide automatic adjustment to any eccentric motion of the disc and/or automatic compensation for seal wear. Drop-tight closure of

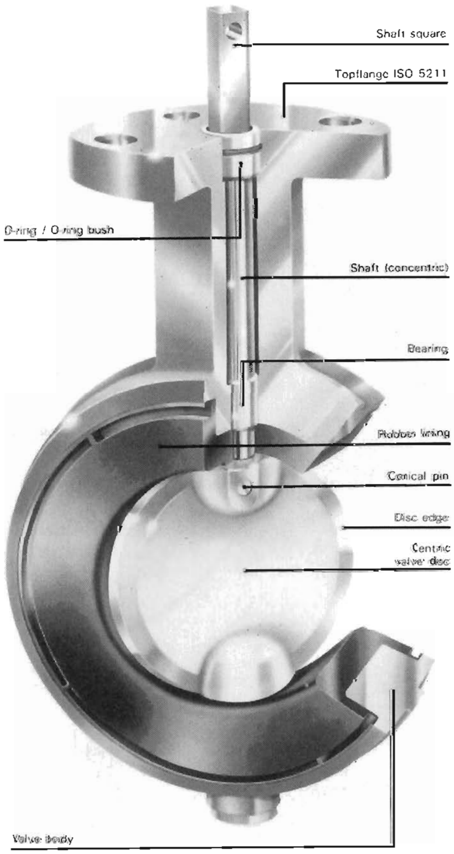


Figure 5. Fully rubber-lined butterfly valve.

any butterfly valve is normal and can be retained for a considerable time before seal replacement is necessary.

Metal seats in butterfly valves have particular advantages and should incorporate the following:

- good tightness—to ANSI B 16.104 Class V and higher
- no jamming
- smooth control—low- and constant-friction load in the control position
- low torque
- high cycle service
- wide temperature range
- good corrosion resistance
- low cavity relief

The metal-to-metal seated valve shown in Figure 7 is designed for bi-directional zero-leakage service. A double-offset (eccentric) shaft together with an offset conical seat achieves the 'camming' action of the disc into the seat. This keeps the sealing ring clear of the seat except in the final shut-off position and provides for non-rubbing rotation.

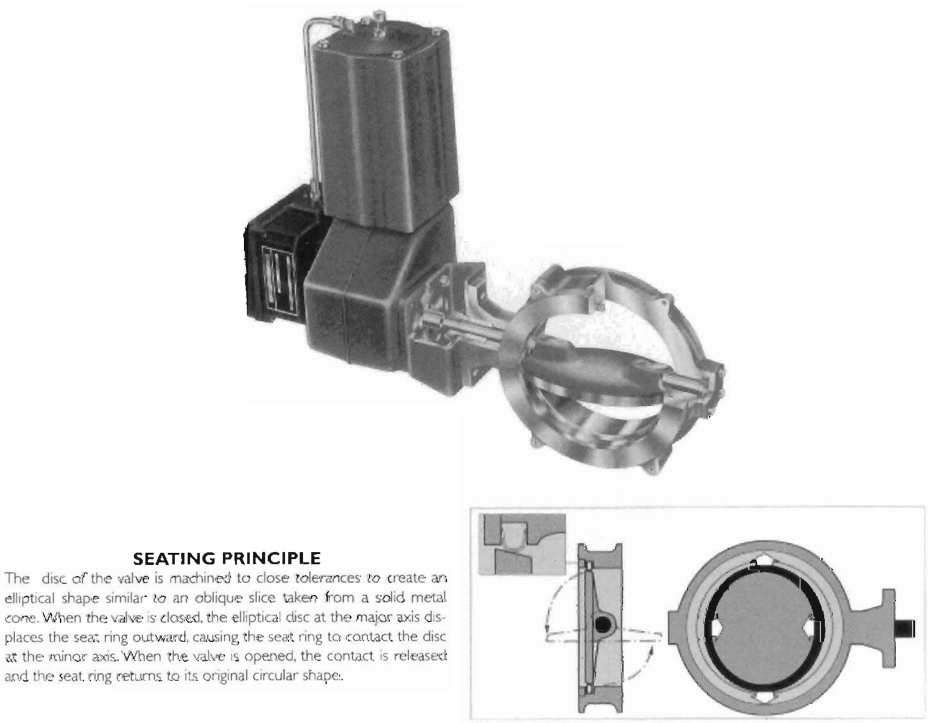
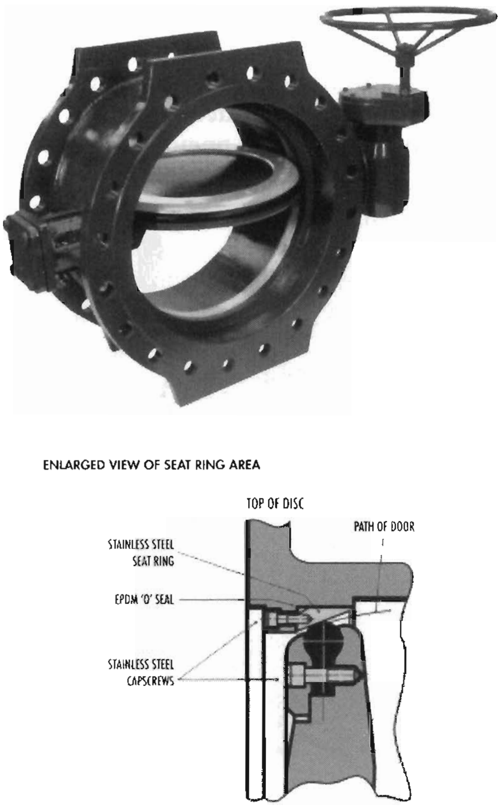


Figure 6. Metal-seated wafer-type butterfly valve.



Stainless steel-seated butterfly valve.

The cone angle allows the seal ring to touch the seat with a contact angle that is uniform and allows a slight amount of wedging.

This type of valve is particularly suited for process and steam duties from cryogenic to high-temperature applications.

Another novel butterfly valve is shown in Figure 8. This valve is designed specifically for dry solids processing. The valve has an inflatable seat that has only 'casual' contact with the valve disc during opening and closing. In operation, air and electric controls operate the valve. They automatically inflate and deflate the seat, allowing a single control input to operate the valve. When supplied with air, the control assembly moves the disc to the closed position and then automatically inflates the seat. When the control signal is received, the seat is instantly deflated and the valve disc moves to the open position. When the control signal is dropped, the assembly returns the disc to the closed position and automatically inflates the seat (Figure 9).

Actuation

All types of control mechanisms can be used to operate butterfly valves—manual by lever or handwheel with reduction gear, electrical by actuator or reduction gear, hydraulic actuator or pneumatic actuator. Choice largely

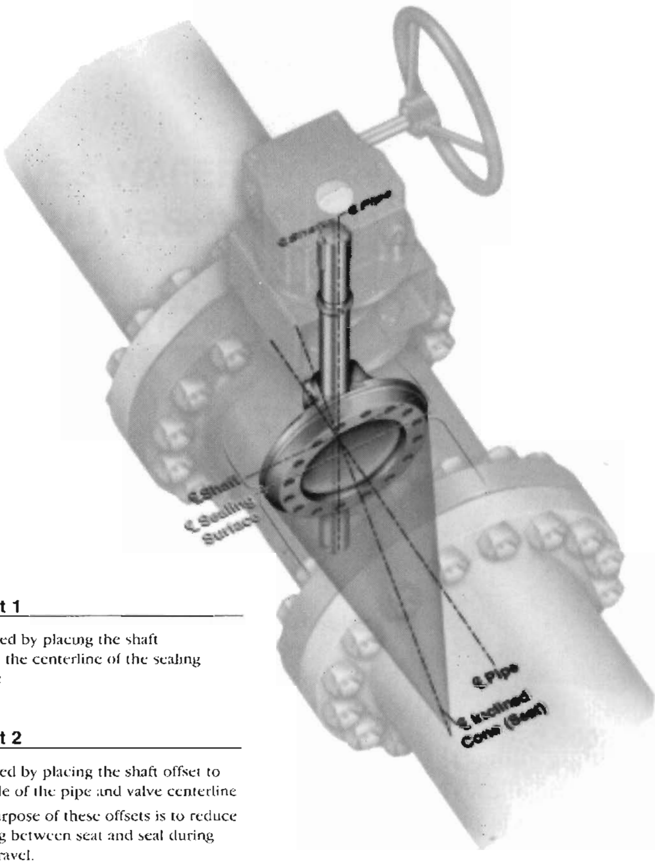


Figure 7. Triple-offset metal-seat butterfly valve.